

Particulate Matter

Week in Review

Week ending 8 August 2026 | 2 August 2026 to 8 August 2026 | 146 records pooled from 7 daily issues

146

RECORDS POOLED
(145 UNIQUE DOIs)

54%

ON THE MEASURE-
MENT
SIDE (79 RECORDS)

10

SUBTOPICS
ACTIVE

30

POOLED EFFECT
ESTIMATES

21

TRIAL REGISTRATIONS
SURFACED

The week in one paragraph

This was the week the low-cost-sensor calibration literature stopped asking whether a correction fits and started asking where it stops working — and the same week the epidemiology independently concluded that source- and co-exposure-resolved metrics beat total mass.

Twenty-four of the 146 pooled records are low-cost-sensor or calibration papers, and the strong ones share a single framing. Ginsburg et al. tested the US EPA PurpleAir correction *worldwide* and found it requires regional recalibration; Zhivkov et al. quantified calibration-transfer error as a function of *distance* from the reference node; Chung et al. characterised PurpleAir deployment *longevity* across the US 2016–2025; Gäbel & Hertig examined *recalibration* practice against sensor ageing. None of these is a new correction equation. All four are about the boundary of an existing one. On the health side the convergence is independent: Cantuaria et al. (*EHP*, **159,769** pooled Danish participants) found that decomposing $PM_{2.5}$ by source inverts which fraction is harmful relative to UFP and elemental carbon (total $PM_{2.5}$ **HR 1.11, 1.05–1.17** per IQR for incident COPD); Hur et al. (**262,288** Kyushu deaths) found the SPM–respiratory-mortality association separates on pollen load (**1.021, 1.005–1.037** on high-pollen days against **1.003, 0.987–1.018** on low); Sassano et al. found household biomass fuel carried **HR 1.20 (1.04–1.37)** for kidney-disease mortality with the estimate splitting on chimney presence. **For this to be a real convergence rather than a reading imposed on it, the two literatures would have to be citing each other. They are not.** Not one of the 146 records validated an instrument against a health endpoint. The convergence is on the *demand* — resolved exposure — and the gap between the two sides is the finding, not the agreement.

Subtopic trend and inflection

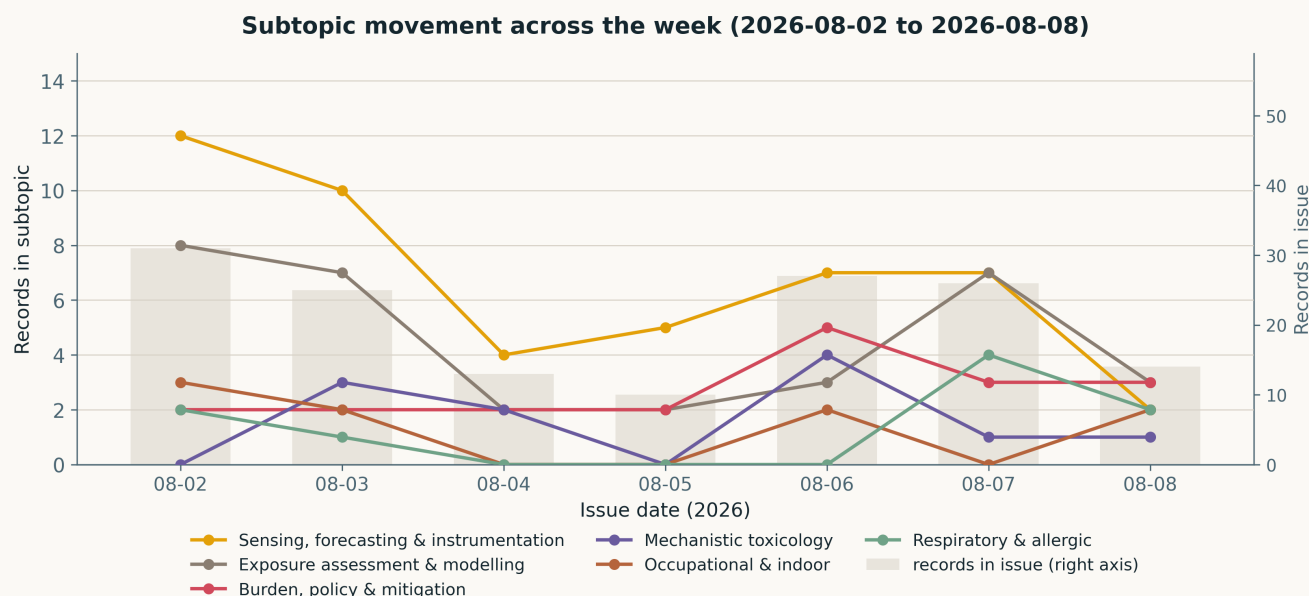


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **Read the denominators first:** the daily totals were 31, 25, 13, 10, 27, 26 and 14, a factor of three between the largest and smallest day, so every count here is a share of a moving base (the 8 August column exists only because `metrics.csv` was rebuilt before this rollout was compiled; a rollout run before its own last daily silently drops that day). The one genuine inflection is 2 August, when the sensing cluster reached 12 of 31 records — and roughly half of that spike is *this pipeline*, not the literature: the Crossref-by-ISSN leg was widened that day and immediately surfaced ten calibration papers that the PubMed legs structurally cannot see. The 4–5 August trough (13 and 10 records) is a mid-week Elsevier deposit lull, and the 8 August floor of 14 is a Saturday. **None of the three is a signal about the field.** The only movement that survives normalisation is the steady 32–54% measurement-side share, which held every single day of the week.

Emerging this week. *Calibration transferability as an object of study in its own right* — absent from the preceding week, and this week carried by four independent groups (Ginsburg, Zhivkov, Chung, Gäbel & Hertig) plus Barkjohn et al.'s Air Sensor Network Analysis Tool, an R-Shiny QA/QC application that is infrastructure rather than a result. *Network design as an optimisation problem* also appears for the first time: Jafari et al. jointly optimise bus-mounted mobile sensing against fixed reference placement, and Bhardwaj & Subramanian pose identifiability-aware source apportionment for sparse city-scale networks. Both treat siting as a decision variable rather than a constraint, which is new in this corpus.

Fading. Neuro and mental-health work collapsed from 22 records in the preceding week to 2, and reproductive/developmental from 24 to 4. **This is almost certainly journal-issue timing, not decline.** The preceding week's count was inflated by a 164-record seed backfill on 26 July that included several review clusters; the honest comparison is 146 records over 7 issues this week against 131 over 6 issues in the preceding week excluding that backfill — 20.9 against 21.8 per issue, i.e. flat. Do not read the subtopic deltas against the raw 295.

Flat but load-bearing. Exposure assessment and modelling (32 records) and mechanistic toxicology (11) held their share through both the peaks and the troughs. Chemical-transport modelling, satellite retrieval and source apportionment are the constant substrate of this corpus; they generate few headlines and most of the methods everything else depends on.

Study architecture

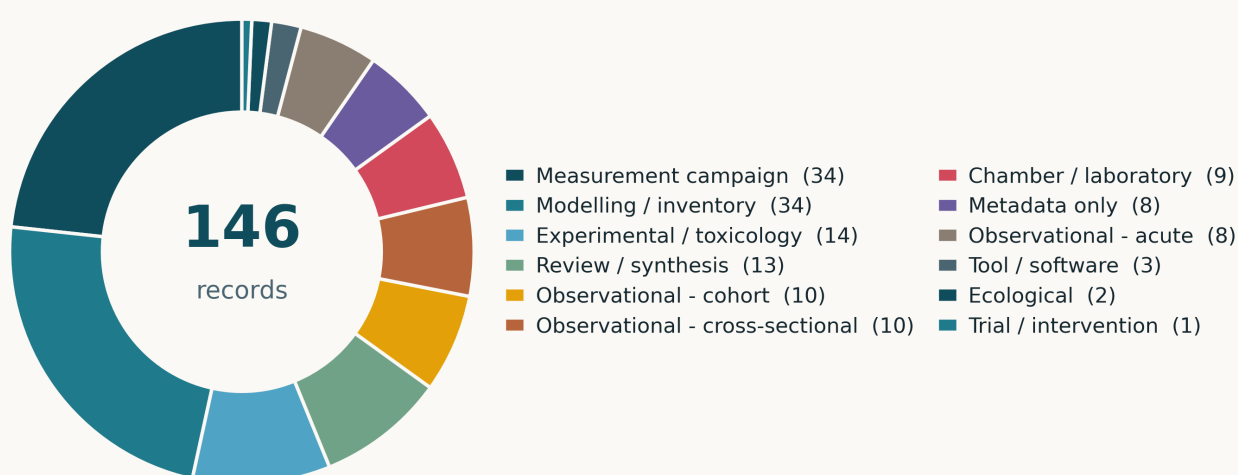


Fig. Pooled study architecture over 146 records. Modelling/inventory and measurement campaigns dominate, which is what an instrumentation-weighted corpus looks like. **Reviews and syntheses are a publication-cycle artefact, not new evidence** — count them separately before reading the corpus as a body of primary work. **The Trial / intervention slice is 1 record in 146** — a fellowship curriculum evaluation with a pre/post knowledge scale, i.e. not a PM intervention at all. Three further records have intervention-adjacent designs that map elsewhere in this grouping (a chamber factorial, a secondary analysis of trial cohort data, a filtration-materials bench study). **Not one randomised intervention on a PM exposure entered the corpus in seven days** — and that is the finding, because 21 such trials are sitting in the registry watch below.

Pooled analytics

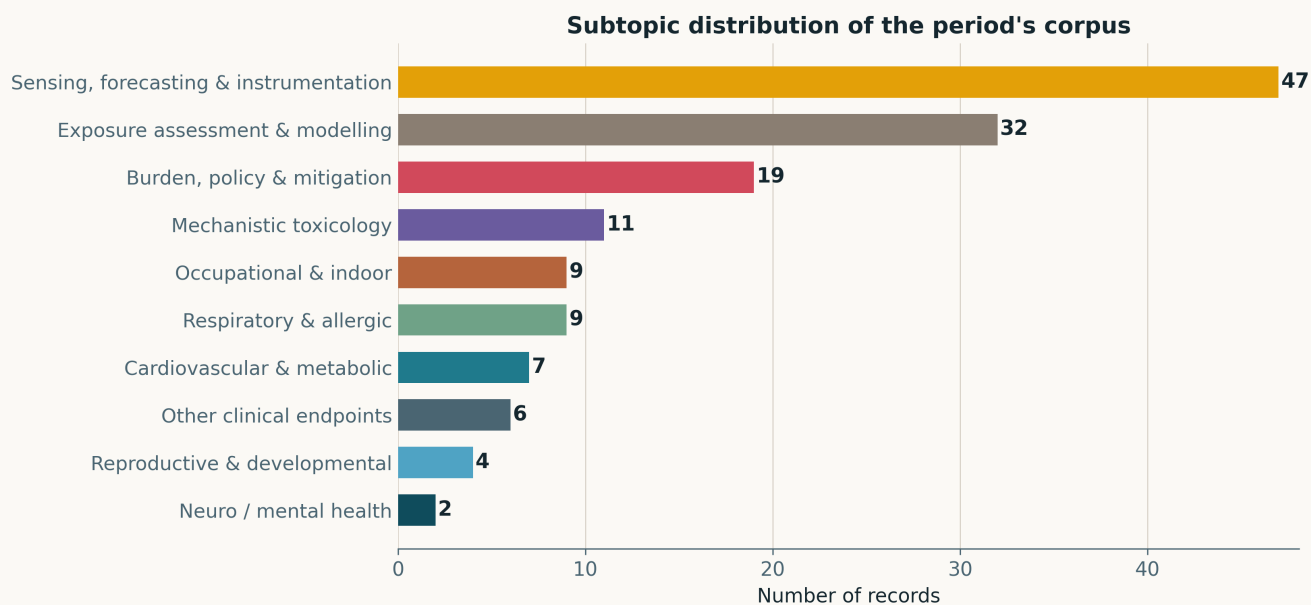


Fig. Pooled subtopic distribution, 2 August 2026 to 8 August 2026. **Sensing, forecasting & instrumentation (47) and Exposure assessment & modelling (32) together account for 79 of 146 records, 54% of the week.** All ten canonical clusters were active, which is a property of a seven-day window rather than of any single day — three clusters were empty on 8 August alone. The clinical tail (Other clinical endpoints 6, Reproductive & developmental 4, Neuro 2) is one to two studies per cluster and carries no weekly trend information.

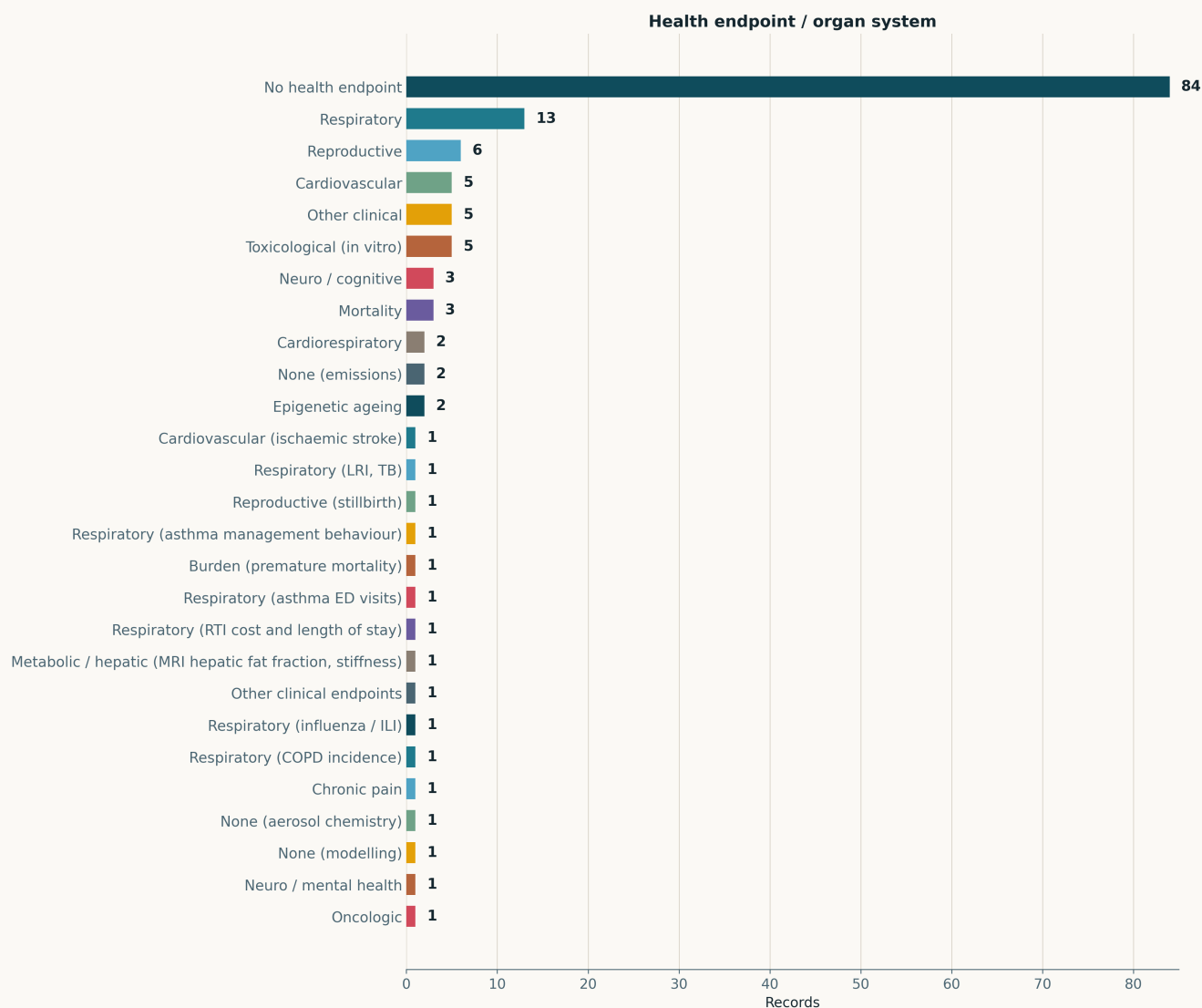


Fig. Pooled health endpoint by organ system. The large no-health-endpoint share is the direct consequence of the 54 % measurement-side weighting above: an instrumentation corpus is mostly records with no human outcome, and that is expected rather than a defect. **Eight records this week were carried on metadata alone** — surfaced with no abstract in Crossref, Europe PMC or Semantic Scholar — and they have no endpoint by construction, not by finding. **Every bar below 3 records is one or two studies**, not a trend; do not read the tail.

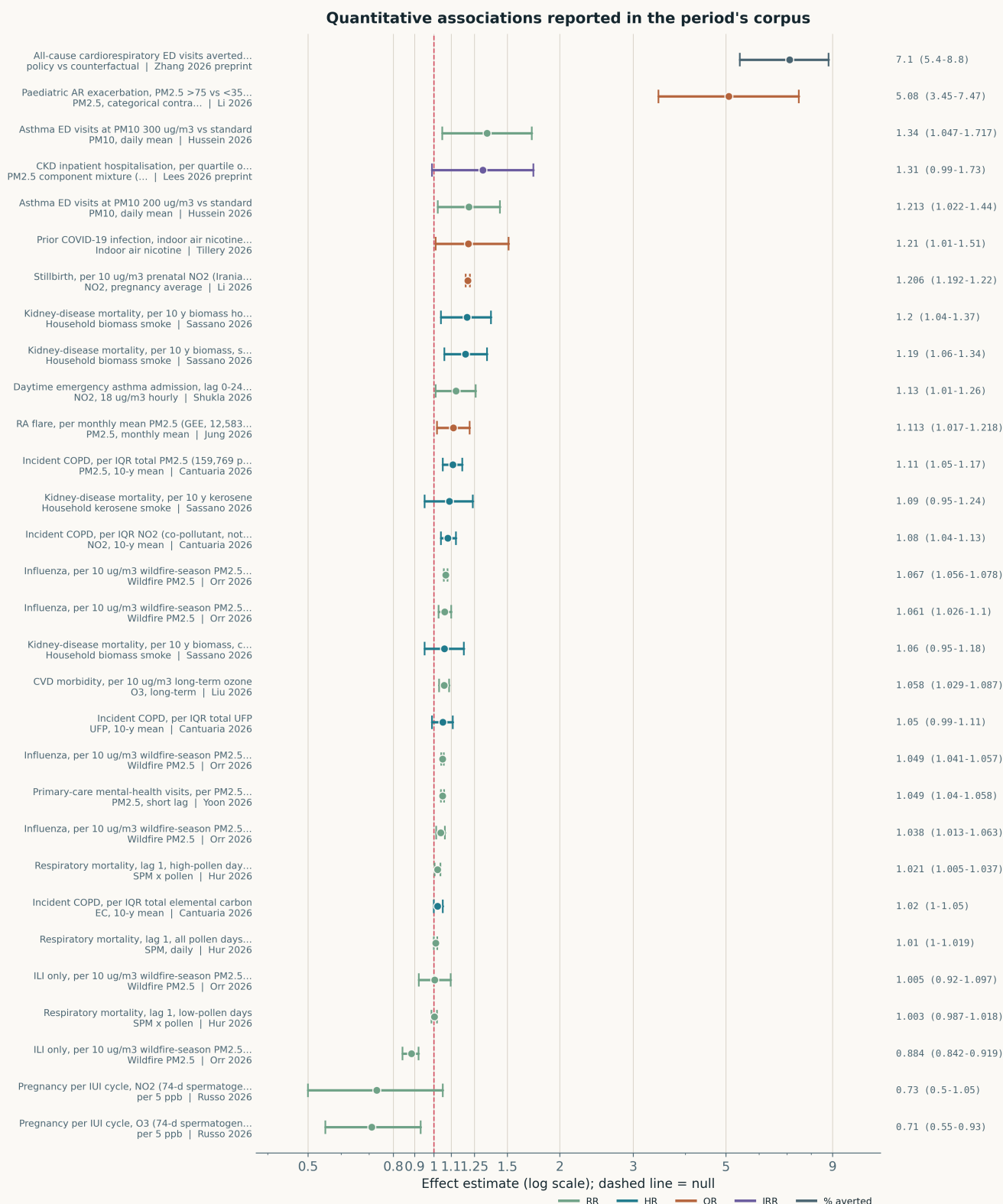


Fig. Pooled effect estimates for the week — read as a caterpillar plot, not a meta-analysis. No summary diamond is drawn and none should be. **Pooling is not defensible here and no I^2 is reported**, for four separate reasons, each sufficient on its own. (i) *Exposure contrasts differ*: per 10 $\mu\text{g}/\text{m}^3$ (Orr), per IQR (Cantuaria), per 10 years of fuel use (Sassano), per binary detection (Tillery), per hourly mean (Shukla). (ii) *Estimands differ*: HR, RR, OR, IRR and one percent-averaged figure share an axis but not an interpretation. (iii) *Records are not independent* — Orr contributes six state-stratified estimates, Cantuaria four source-resolved ones, Hur three pollen strata and Sassano four ventilation strata; four papers supply 17 of the 30 points. (iv) *Not all exposures are PM*: Li et al. and Liu et al. are explicitly flagged co-pollutant in the corpus, and Shukla and Tillery are NO_2 and nicotine. **The one subset that is genuinely comparable** is Orr's six wildfire- $\text{PM}_{2.5}$ -influenza estimates, all per 10 $\mu\text{g}/\text{m}^3$ on the same outcome definition — and they range from **0.884 (0.842–0.919)** to **1.067 (1.056–1.078)** with non-overlapping intervals, which is the week's most important heterogeneity precisely because it is *within* a single paper and tracks the outcome-ascertainment boundary, not geography.

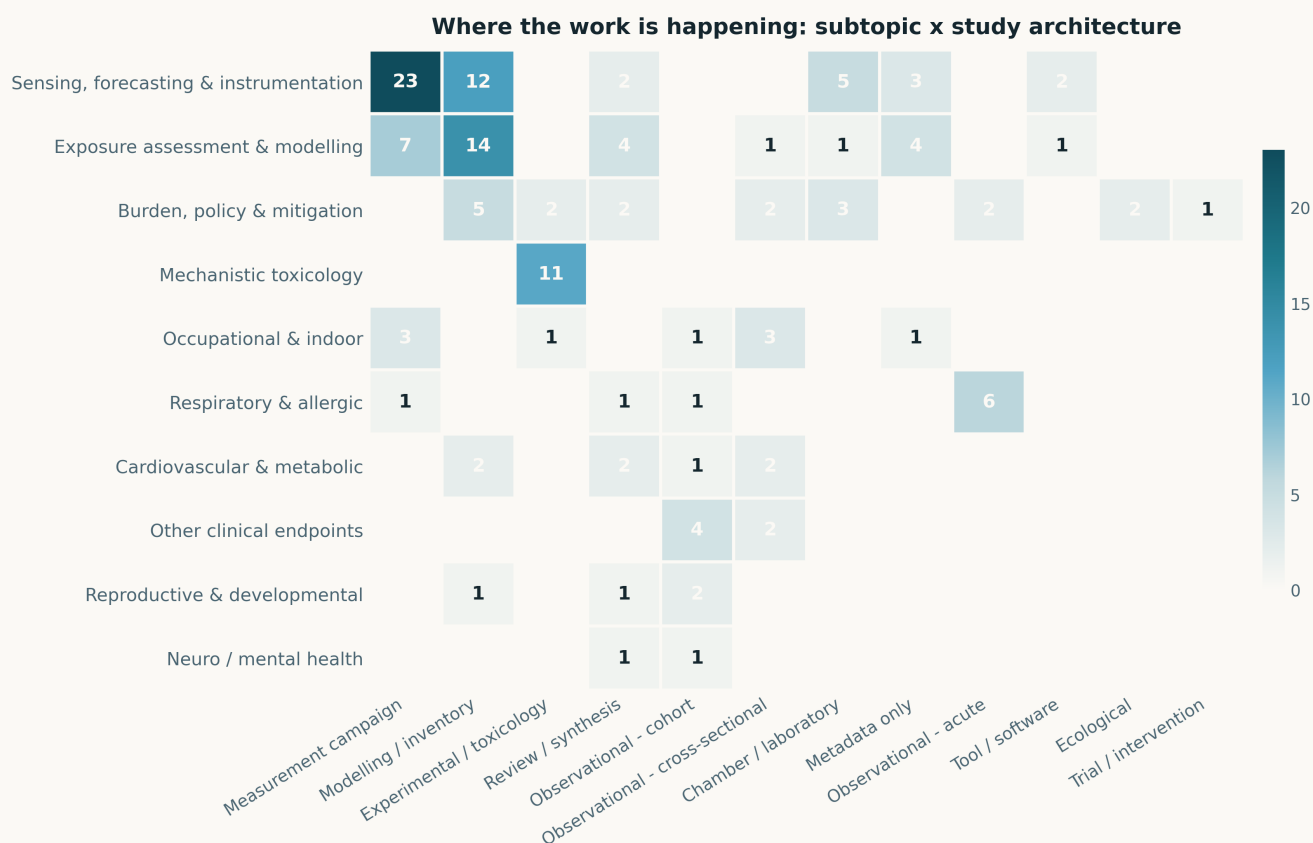


Fig. Pooled subtopic $\times$ study-architecture cross-tabulation over the full week. **The cells that stay empty are the point.** At $n = 146$ — seven times a daily corpus — the *Sensing* and *Exposure assessment* rows still contain no *Trial / intervention* cell and no cell that couples an instrument to a health outcome. *Mechanistic toxicology* remains entirely experimental with no bridge to human biomarker work, the same structural gap this watch recorded on 27 July. A week of data does not close a cell that a day of data left empty; it establishes that the cell is empty by construction of the field, not by sampling.

Groups, venues and most-cited work

Recurring author and group clusters. The naive first-author count is **misleading and must not be used**: the top entries are Li (5), Wang (4), Zhang (3), Liu, Zhao, Guo, Yang and Kim (2 each), which are among the most common surnames in the world and almost certainly denote distinct groups. Two repeats survive inspection. **Orr** appears twice in genuinely different roles — the *EHP* wildfire-smoke and influenza analysis across six US states, and a *JMIR* feasibility study of an in-home air-quality monitor in families of children with asthma — and those two records are the closest thing this week has to the instrument–endpoint bridge that the heatmap says is missing. **Kumpika** appears twice because **the same paper shipped in two issues**: DOI 10.1080/26395940.2026.2616098 was carried on 2 August as “PMS7003 light-scattering sensors” and again on 3 August under a broader title. **This is a deduplication failure, one record in 146, and it is stated here rather than quietly corrected** — the pooled count is 146 records over 145 unique DOIs. Beyond that, repeat appearances in this corpus are the exception: 146 records, no institution with more than two, and no evidence of a single programmatic output stream inflating apparent consistency.

Venue concentration. *Atmospheric Chemistry and Physics* is the largest single contributor at **11 records**, and that is a themed concentration rather than a research surge — ACP publishes continuously and the deposits cluster by editorial batch. After normalising for a label defect described below: *Environmental Pollution* 8, *Atmospheric Environment* 7, *Research Square* (preprint) 6, *Atmospheric Measurement Techniques* 5, *Journal of Hazardous Materials* 5, *Environmental Monitoring and Assessment* 5, *Atmospheric Pollution Research* 4, *J Air & Waste Management Association* 4, *Environmental Health Perspectives* 3. **Defect worth recording: journal names are stored as free text and the same venue appears under two spellings** — Atmos Environ (4) and Atmospheric Environment (3), Environ Pollut (5) and Environmental Pollution (3), Res Sq (2) and Research Square (4). Uncorrected, the register understates venue concentration by roughly a third at the top of the list.

Most-cited work of the period. Not reportable, and ranking it would be noise. Every record in this corpus entered a PubMed or Crossref index within the last seven days; citation counts are zero or near-zero for all 146 and carry no signal about importance. Consensus returns citation counts, but for the records it surfaced this week those counts belong to 2019–2025 papers that were *not* in the entry window and were rejected for exactly that reason. A defensible most-cited list requires a lag of at least 18 months and will be reported in the yearly issue, not weekly.

Trial registry watch

From `state/trials.json`: registrations surfaced inside the window. Daily issues accumulate; the rollout is where they surface. **21 registrations** entered state during 2–8 August — 16 from the 2 August sweep, 1 on 4 August, 3 on 7 August and 1 on 8 August. Note a hard limitation of the connector: it exposes *no* last-update-date filter, so these are relevance sweeps, not true window queries, and `first_seen` means “first seen by this pipeline”, not “registered or updated on that date”.

The structural finding first. Of the 21, at least 13 are sham- or placebo-controlled portable-air-cleaner or filtration trials. That is the intervention literature in this field almost in its entirety, and it has a common design problem: a HEPA unit in a home typically buys a 30–60 % reduction in indoor PM_{2.5} over a few months, which is a small absolute exposure contrast, and the trials are then powered on **surrogate endpoints** — blood pressure, spirometry, biomarkers, cognitive scores — rather than events. Where a hard outcome is attempted, enrolment is far too small for it.

Adequately powered for their stated endpoint (surrogate). **NCT06376994** (JHSPH Center for Clinical Trials, air cleaner vs sham in COPD, **n=770**, primary completion 2029-08-31) and **NCT05874479** (NYU Langone, active vs sham PAC in NYC public housing, blood pressure, **n=440**, 2027-09) are the two that can plausibly detect a several-mmHg or modest spirometric difference. **NCT06070428** (Henry Ford, PATHFINDER, PAC at discharge in heart failure, **n=400**, 2027-06) is the only one of the three with an event-adjacent population, and even there the endpoint is not a hard event.

Underpowered or endpoint-mismatched — flagged. **NCT06534385** (Alaska Native Tribal Health Consortium, HEPA purifiers in rural Alaska homes, **n=15**) and **NCT07118189** (Kaiser Permanente, N95 + indoor air cleaner for wildfire smoke, **n=20**) cannot support any inferential health claim and should be read as feasibility work. **NCT07536178** (UBC, controlled lodgepole-pine woodsmoke vs filtered air, **n=24**) is a mechanistic exposure study, correctly small, and is the cleanest causal design in the list. **NCT06197477** (Hasselt, school purifiers, **n=900**) is the largest trial in state and carries **four cognitive co-primary endpoints with no secondaries and no stated multiplicity control** — with status UNKNOWN past a December 2025 primary completion, it is now also a publication-risk item.

New on 8 August. **NCT05368493** (New York State Psychiatric Institute, prenatal air pollution and neurodevelopment in Puerto Rican children, **n=182**, ENROLLING_BY_INVITATION, primary completion 2028-06-01). `analyze_endpoints` returns a *single* primary — bilateral middle frontal cortex and corpus callosum volumes, scanned at either 6–36 months or 6–11 years — and **zero secondary endpoints**. Two problems worth recording now rather than at readout: the registry classifies it INTERVENTIONAL when the only listed intervention is an MRI task, so it is an observational neuroimaging cohort mislabelled; and the primary is measured at two non-overlapping developmental windows depending on enrolment timing, which makes the effective sample for either window well under 182. Exposure is modelled ambient PM, not measured.

Standing watch list — results due or overdue. **NCT05994937** (NYU Langone, PAC glycaemic/cardiometabolic biomarkers, **n=150**) reached primary completion **2026-03-16** with **no results posted** — five months overdue. **NCT06534385** (2026-04-01) and **NCT07047430** (U. Washington, airport-adjacent asthma, 2026-06-01) are also past completion. **NCT07111208** (MARKOPOLO, noise and UFP, **n=180**) had primary completion **2026-07** and is the only UFP-exposure interventional registration in state. **NCT07344987** (Peking Union, ILD crossover, 2026-12) and **NCT07118189** (2026-12-31) close this year.

Open gaps

- ▶ **Measurement–epidemiology coupling: still zero.** Across 146 records and 24 low-cost-sensor papers, **not one validated an instrument against a health endpoint.** The nearest approach is Orr et al. (*JMIR*), which measured feasibility and acceptability of an in-home monitor in families of children with asthma but reported no exposure–response. The two literatures were adjacent all week and never touched.
- ▶ **Ultrafine and speciated metrics.** Five records mention UFP; **exactly one measures it as an exposure** (Cantuaria et al., where the local-traffic fraction dominates for UFP and elemental carbon while the *non*-traffic fraction dominates for PM_{2.5} — an inversion with direct siting implications). The rest are a narrative review, a dosimetry update, a filtration-CFD paper and a mechanistic pathway study. Against 55 PM_{2.5}-only and 18 joint PM_{2.5}/PM₁₀ records, the deposition physics remains badly under-served.
- ▶ **Geographic coverage vs. attributable burden.** China 21, North America ≈24, Europe ≈10 against **South Asia ≈9 and sub-Saharan Africa 5.** India and Bangladesh together carry a larger share of global PM-attributable DALYs than China and North America combined and contributed fewer than ten records in seven days. The one sub-Saharan primary instrumentation record — Rehbein et al. on source triangulation from a low-cost network in Ile-Ife — is exactly the kind of work the gap needs, and it is one paper.
- ▶ **Calibration transferability: improving, and now the week’s strongest theme.** Unlike previous weeks, this corpus *does* contain held-out and out-of-region evaluation: Ginsburg et al. (US correction applied worldwide), Zhivkov et al. (error as a function of distance from reference), Gomez et al. (ML/DL against a T640X reference), Chung et al. (deployment longevity, US 2016–2025). **What is still missing is a common benchmark** — every one of these evaluates on its own network, so the numbers are not comparable across papers.
- ▶ **Negative and null results.** Present, which is a good sign. Russo et al. report **protective** point estimates in a spermatogenesis-window analysis (RR 0.71, 0.55–0.93 and 0.73, 0.50–1.05) that are more likely selection or window misspecification than a real benefit; Orr et al. report a protective estimate in Washington state (0.884, 0.842–0.919) that the paper itself cannot explain except by outcome misclassification; Cantuaria reports two null source-resolved HRs (1.02, 1.00–1.05 and 1.05, 0.99–1.11). **A corpus with no nulls is a reporting-bias signal; this one has them, and they are informative rather than decorative.**

Where this is heading — judgement, clearly marked as such

The methodological demand the field is converging on is a resolved exposure record, and the convergence is arriving from two directions that are not talking to each other. From epidemiology: source apportionment (Cantuaria), co-pollutant interaction (Hur), microenvironment (Sassano, Tillery) and sub-daily lag structure (Shukla) each independently showed this week that a scalar mass concentration is not a sufficient statistic for the health-relevant exposure. From instrumentation: four groups simultaneously stopped publishing new correction equations and started publishing the *failure boundaries* of existing ones. Both are asking for the same object. Neither is building it. Confidence: moderate-to-high on the description, low on any claim that this week caused it — seven days is not enough to establish a turn, and 2 August’s calibration cluster was partly an artefact of widening this pipeline’s own Crossref leg.

What would change the picture. A single paper that deploys a calibrated low-cost network and reports an exposure–response against a clinical endpoint would collapse the empty heatmap cell that has now survived two weeks of accumulation. Nothing in the 21 registered trials will supply it: they are almost all filtration interventions with surrogate endpoints, and the two that could (NCT06376994, n=770; NCT05874479, n=440) read out in 2027 and 2029. If the coupling comes, it will come from a cohort retrofitting a dense sensor network onto existing outcome ascertainment, not from a trial.

What a distributed low-cost PM sensing programme should do differently. Three concrete things this week’s corpus argues for. (1) Report calibration performance as a function of distance and time from the co-location reference, not as a single held-out R^2 — Zhivkov et al. give the framing and Chung et al. give the longevity data to do it. (2) Instrument the co-exposure, not just the mass: Hur’s pollen interaction and Cantuaria’s source split both say the node needs a second channel (fluorescence, coarse-mode, or a traffic-proxy gas) before its mass output is epidemiologically interpretable. (3) Resolve below 24 hours and keep it — Shukla localises the asthma signal at a 3–6 hour lag, which no filter-based reference method can see and which is the one regime where a low-cost network is not merely cheaper but strictly better. **The week was informative, and unusually so on the instrumentation axis specifically.**

Paper-level digest — the week by subtopic

Every record of the period, grouped by subtopic. **Two tiers, and the difference matters.** Entries carrying a full critical summary are lifted *verbatim* from the archived daily issue that first published them — they were read, and the stated weaknesses are assessed weaknesses. The remainder appear as precis or metadata-only entries in the lighter style: those records were surfaced and classified but **not read**, either because no abstract was available in Crossref, Europe PMC or Semantic Scholar (8 records this week) or because they entered on metadata alone. **Absence of criticism in the lighter tier means the record is unassessed, not that it is sound.** Do not cite a precis entry as though it had been evaluated.

Sensing, forecasting & instrumentation

47 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (47 OF 47 RECORDS IN THIS SUBTOPIC)

Adeniji et al. 2026 PLOS Glob Public Health

Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria

Hourly PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ over 2012–2023 with six meteorological covariates. **PM_{2.5} and PM₁₀ dominate and persistently exceed WHO guidelines**, peaking in the December–March Harmattan; regression against temperature gives R² 0.48–0.52 with negative coefficients on rainfall and humidity, consistent with thermal stagnation and wet scavenging. The aggregate AQI sits at 39–51 (“moderate”) while the health-risk-adjusted index exceeds 90 seasonally — **a 2-fold divergence between the two indices at identical concentrations**, which is the paper’s real contribution and an argument for HAQI-style reporting on any network serving a PM-dominated airshed. Single station for a metropolitan area, and the provenance and calibration history of a 12-year record in a data-sparse setting are not documented.

doi: 10.1371/journal.pgph.0006683

Astaneh et al. 2026 Atmos Environ

Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions

METADATA ONLY. Using source-tagged particulate tracers as features in an ML bias-correction layer over a photochemical model is a transferable construction even though the target here is ozone, not PM.

doi: 10.1016/j.atmosenv.2026.122288

Barkjohn et al. 2026 Atmosphere

An R-Shiny application for sensor-versus-reference QA/QC

Tool paper describing an application that lets air-quality agencies (i) assess sensor data quality against nearby regulatory reference monitors, (ii) apply baseline QA/QC and (iii) characterise local conditions, with reproducibility as an explicit design goal. Demonstrated on two cases: PM_{2.5} sensor performance under Midwestern wildfire smoke, and a year of O₃ sensor performance. The authors are explicit that advanced correction remains out of scope. **Limitation: no new method and no quantitative evaluation of the tool itself** — the case studies illustrate rather than validate. Carried because availability, not novelty, is what determines whether a small group’s sensor data ever gets screened at all; this lowers that barrier and standardises the first pass.

doi: 10.3390/atmos16111270

Büttner et al. 2026 Atmos Meas Tech

PIA v3.0.0: automated quality assurance for PINE ice-nucleating-particle data

Software description for the analysis chain behind the Portable Ice Nucleation Experiment. Two components documented in detail: automated data quality control, and the algorithm separating aerosols and droplets from ice crystals by optical size. The second half gives quality-assurance recommendations for PINE operation. Not a PM_{2.5} paper, but the most directly transferable artefact in the issue — a published, versioned QA/QC specification for a continuously operating particle instrument is precisely what distributed sensing networks lack.

doi: 10.5194/amt-19-5007-2026

Burke et al. 2026 Atmos Environ

Polar-grid nonparametric trajectory analysis recovers near-field source structure

Nonparametric trajectory analysis combines continuous concentrations with local-scale back trajectories from wind speed and direction to map source-area impact. The regular Cartesian grid plus kernel smoothing used previously suppresses spatial variability exactly where a near-source monitor needs it; re-gridding onto polar coordinates gives finer resolution close to the site. Tested on one year of continuous data from a near-road network site in St. Louis. Areas nearest the roadway contributed most (1.4 µg/m³ BC, 39 ppb NO_x), and **polar NTA raised maximum estimated contributions by 22% (BC) and 39% (NO_x)**. Background-subtracted PM_{2.5} tracked the traffic-related species but also revealed nearby point-source contributions. **Limitation: one site, one year, no independent validation** of the recovered source field — the improvement is demonstrated as internal consistency, not against known emissions. Immediately relevant to siting and to interpreting single-node data in a sparse network.

doi: 10.1016/j.atmosenv.2026.121951

Chung et al. 2026 Sensors

PurpleAir deployment trends, longevity and uncertainty across the United States

Every publicly shared outdoor PurpleAir unit, 2016–2025, with deployment duration taken as the span between first and last public observation, summarised nationally, by Census region and by state. **Most sensors persist beyond three years**; density and longevity are highest in the western US. Descriptive PM_{2.5} summaries for California, Minnesota, Pennsylvania and Texas by deployment duration and urban–rural class. The limitation is that it characterises *availability*, not accuracy — persistence of a data stream is not persistence of calibration, and the paper explicitly does not address drift. That is exactly the join left open against Kumpika et al. below.

doi: 10.3390/s26061789

Chyzykhov et al. 2026 J Hazard Mater*Particle-bound water and the composition dependence of gravimetric PM mass*

Karl Fischer coulometric titration with stepwise heating on 35 single-component chemicals and CRMs at two filter loadings (0.6 and 2.0 mg, summer- and winter-like), plus a compositionally realistic artificial PM mixture built both by sequential addition and in bulk. Deliquescent inorganic salts retain the most water; metals, metal oxides, PAHs and graphitic carbons sit near background. Water-to-solute molar ratio falls as absolute loading rises — compacted powder is less vapour-accessible — and temperature-resolved release clusters into three material groups differing mainly in strongly bound (250 °C) water. Bulk artificial-PM release profiles matched ambient PM₁ filters. **Limitation: powders are not ambient particles**; mixing state, morphology and coating history are not reproduced, so the transfer to real filters is argued from profile similarity, not demonstrated. For distributed sensing this is the uncomfortable upstream result — the mass reference itself carries a composition-dependent water term, so an empirical RH correction fitted against it is partly correcting the reference.

doi: 10.1016/j.jhazmat.2026.143142

Colarusso et al. 2026 Internet of Things*Mobile urban sensing: spatio-temporal observability analysis and optimisation*

Fleet-aware framework for hybrid mobile-fixed sensing on an Edge-Fog-Cloud architecture, evaluated on real data from three sensing vehicles covering 11.37 km² in Benevento, Italy. Introduces spatial and temporal observability indices and a value-of-fleet metric, then uses spatio-temporal interpolation plus what-if analysis to quantify how fleet size, route redundancy, sampling density and data loss propagate into reconstructed-map accuracy; a greedy co-design procedure places fixed nodes against the residual. Limitation: a single city, a single small fleet, and interpolation-based ground truth — the observability indices are internally consistent but not externally validated. The framing is the transferable part: opportunistic mobile coverage becomes a measurable quantity rather than a qualitative claim.

doi: 10.1016/j.iot.2026.101907

DeMarsh et al. 2026 (backfill) Atmosphere*Community PurpleAir network versus regulatory monitors for PM_{2.5} in Fresno County*

The SJV-CAIR network at UC Merced, built with community-based organisations, compared against regulatory monitoring across Fresno County for June–July 2023. Community monitors returned **concentration estimates comparable to regulatory instruments** where the two co-locate, but identified **four weeks in which some locations episodically exceeded the NAAQS that the regulatory network did not flag**, and interpolation maps built with the denser network resolved spatial structure the regulatory-only map did not. Two summer months in one county is a short and seasonally atypical window for the San Joaquin Valley, and no formal co-location correction is reported. Carried as backfill — Crossref created 11 February 2026, outside this entry window — and must be excluded from weekly trend lines.

doi: 10.3390/atmos17020187

Ginsburg et al. 2026 Aerosol Sci Technol*Global and regional calibration of low-cost PM_{2.5} sensors (PurpleAir)*

1,013 PurpleAir units co-located within 500 m of reference monitors on six continents, Jan 2023–Mar 2025, assessed at 1-h and 24-h averaging by R^2 , RMSE, nRMSE, precision (CV) and AQI category correctness. Raw readings overestimate PM_{2.5} in most regions and miss EPA performance targets; US-derived corrections show limited transferability outside North America; a data-driven global quadratic helps in Australia/NZ and hurts elsewhere; **regional equations meet EPA 24-h thresholds in multiple regions and lift 24-h AQI correctness to ≥77%**. Limitation: co-location proximity (≤500 m) is a proxy for comparability, so some regional “bias” is spatial-gradient error, and continents are represented very unevenly. Still the strongest available empirical case that LCS network calibration must be local.

doi: 10.1080/02786826.2026.2676293

Gómez et al. 2026 Sensors*Machine- and deep-learning calibration of low-cost PM₁₀/PM_{2.5} nodes*

LCS nodes co-located with a T640X reference at Universidad Distrital, Bogotá, with RH, temperature and absorption flow from an Automet station. Pipeline: filtering, segmentation, **FastDTW** alignment, then six calibration models (two statistical, three ML, one DL). Random Forest and XGBoost performed best; the authors argue that the FastDTW preprocessing step and the inclusion of absorption flow — not the model choice — carry the accuracy gain. Single-site co-location with no transfer test to an uncalibrated node, so generalisation is unestablished.

doi: 10.3390/s26030796

Guo et al. 2026 ACS ES&T Air*Mechanistic RH correction by separating aerosol liquid water from dry mass*

Two-stage framework for a nephelometer plus optical particle counter: (i) Mie-theory optical calibration accounting for size distribution, refractive index and hygroscopic growth; (ii) quantification of aerosol liquid water mass from EPA Chemical Speciation Network composition to recover dry-equivalent PM_{2.5} at high RH. Corrected values agree closely with EPA reference measurements, and the recovered ALW is itself useful for aqueous-phase aerosol chemistry. Practical guidance is given for sites without co-located CSN speciation. **Limitation: the method is only as good as the speciation surrogate** — outside the CSN footprint the ALW term becomes an assumption, and the paper does not report how much accuracy is lost when the surrogate is wrong. This is the right-shaped answer to the humidity problem for anyone unwilling to refit an empirical curve at every new site.

doi: 10.1021/acsestair.5c00225

Gäbel & Hertig 2026 Atmos Meas Tech*Recalibration of low-cost O₃ and PM_{2.5} sensors: linking practices to recent air sensor test protocols*

Yearlong urban-background co-location at Augsburg of Alphasense OX-B431 and Sensirion SPS30 against reference instruments, spanning −10 to 36 °C, 19–96% RH and PM_{2.5} to 153 µg m^{−3}, with MLR, ridge, random forest and XGBoost calibration functions evaluated against the current US EPA and CEN air-sensor test protocols. A five-month baseline training gives PM_{2.5} R^2 0.93–0.97 in training and **0.84–0.93 in the following test period**, RMSE 2.04–3.02 µg m^{−3} — the decay is the finding. Testing pairwise recalibration cycles, **monthly in-season recalibration** was required to keep uncertainty inside the data-quality objective for indicative measurements. Limitation: one site, one year, two sensor models, so the cadence is an existence result rather than a general rule. **Out-of-window backfill** (created 19 Feb 2026); *AMT* is not indexed by PubMed. Exclude from weekly trend lines.

doi: 10.5194/amt-19-1293-2026

Huang et al. 2026 Atmos Pollut Res*Metal ions and pH as controls on the optical properties of atmospheric brown carbon*

Crossref deposit of 7 August, **metadata only — no abstract retrievable**, so no summary is offered. Logged because brown-carbon absorption is the aerosol optical property that most directly confounds the fixed refractive-index assumptions built into low-cost optical particle counters.

doi: 10.1016/j.apr.2026.103167

Jafari et al. 2026 IEEE Open J Intell Transp Syst

Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring

Mathematical-programming formulation selecting simultaneously **which buses to instrument with low-cost sensors and where to place reference stations**, maximising spatiotemporal network coverage under a budget constraint while requiring that every mobile sensor pass a reference site frequently enough to be recalibrated. Demonstrated on a Rotterdam case study covering residential and industrial areas. The contribution is structural rather than empirical: recalibration cadence enters the siting objective as a constraint, so coverage and data quality are traded off in one problem instead of sequentially. Limitation: coverage is geometric — optimised against route and population geometry, with no validation against a measured concentration field, and no treatment of the sampling bias that fixed bus routes impose. This is the record held on 4 August and recovered on 5 August via Semantic Scholar.

doi: 10.1109/ojits.2026.3706855

Jin et al. 2026 Atmos Chem Phys

Emissions, chemistry and health impact of aged wildfire smoke over Missoula, Montana

Hourly surface observations of PM_{2.5}, CO, NO_x, O₃ and **75 speciated VOCs** through a strong 2020 smoke episode, used as a direct test of how wildfire emissions, ageing chemistry and health effects are represented in GEOS-Chem. Smoke three or more days old, transported from California and the Pacific Northwest, raised CO, PM_{2.5} and total measured VOCs by **factors of 2–8**, with hourly maxima of 800 ppb CO and **120 µg/m³ PM_{2.5}**. The value here is the speciated VOC suite alongside mass: it is the composition record against which a mass-only low-cost network's health interpretation has to be judged, and it is one city and one episode, so the transport chemistry is not generalisable without replication at other receptor sites.

doi: 10.5194/acp-26-11047-2026

Kumpika et al. 2026 Environ Pollut Bioavail

Humidity and ageing corrected together, and duty cycle as a drift control

PMS7003 units in Chiang Mai across varying RH, validated against a government BAM and verified in dry air with a diffusion dryer. Polynomial corrections up to third order gave strong agreement at elevated humidity ($R^2 = 0.9737$). The distinctive result is operational rather than statistical: **limiting operation to 20 minutes/day measurably reduced long-term performance drift** relative to continuous running. **Limitations: single site, single sensor model, single aerosol regime** (Chiang Mai's biomass-burning season is compositionally atypical), and the duty-cycle finding is confounded with total particle throughput — a sensor that samples less sees less mass through its optics, so “ageing” and “fouling” are not separated. Worth replicating; if it holds, duty cycling is a nearly free lifetime extension for battery-constrained nodes.

doi: 10.1080/26395940.2026.2616098

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doi: 10.1080/26395940.2026.2616098

Lei et al. 2026 Atmos Chem Phys

Eight-year speciated PM₁ record in Dublin: heating emissions versus transboundary transport

Continuous chemically speciated PM₁, 2016–2023, resolving two distinct pollution regimes: short intense events (hours, PM₁ >100 µg/m³) driven by local solid-fuel home heating, and moderate multi-day events (<60 µg/m³) from continental transport. Annual PM₁ fell from 6.5 to <5.0 µg/m³, led by nitrate and ammonium (−0.11 and −0.09 µg m^{−3} yr^{−1}). **Transported oxygenated organic aerosol rose (+0.34 µg m^{−3} yr^{−1})** while locally formed OOA barely fell, so the composition shifted even as mass declined — and 2023 showed a heating-driven rebound. A demonstration that a mass-trend headline can be exactly wrong about what a network is now measuring.

doi: 10.5194/acp-26-10927-2026

Li et al. 2026 Atmos Chem Phys

Summertime ice-nucleating particle concentrations over the Eurasian Arctic seas

Shipborne immersion-mode INP measurements from the Arctic Century Expedition across the Barents, Kara and Laptev Seas and adjacent high-Arctic islands, August–September 2021. N_{INP} was lower than at continental high-latitude sites, most clearly below −15 °C, implying sparse mineral-dust INP; concentrations peaked in ice-free water, marginal ice zones and near land, and were very low inside the ice pack, with the maximum north of Novaya Zemlya under back-trajectories from the western Siberian coast. The authors conclude sources are local to regional with little long-range transport signature. Limitation: one summer and one cruise track, so the seasonal and interannual envelope is unconstrained. Peripheral to PM exposure, but relevant to interpreting particle counts in clean marine air, where the concentration floor is set by local sources rather than by transport.

doi: 10.5194/acp-26-10835-2026

Li et al. 2026 Front Pediatr

Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical and ambient data

Development cohort $n = 2,000$ linking NHANES (2005–06, 2011–18) to EPA AQS and NOAA data; external cohort $n = 50$ from Wuhan. L2-regularised logistic regression was prespecified and beat random forest, XGBoost, LightGBM and SVM under nested 5-fold CV. Internal AUC 0.81 (0.74–0.88), external 0.71 (0.50–0.92). Baseline symptom score, same-day PM_{2.5} and total IgE dominate SHAP; **PM_{2.5} >75 vs <35 µg/m³ gives OR 5.08 (3.45–7.47)**. The external cohort is 50 children on a different continent — the authors say plainly this is preliminary transportability, not external validity, and they are right.

doi: 10.3389/fped.2026.1865399

Liu et al. 2026 Atmos Chem Phys

From κ to χ : particle-resolved evaluation of hygroscopicity-based mixing-state retrieval

PartMC-MOSAIC urban-plume ensembles spanning emissions, ageing and meteorology were used to compare the particle-resolved reference mixing-state index against the Yuan–Zhao retrieval from HTDMA κ distributions, per diameter over 50–250 nm. The retrieval performs well for many populations but **systematically overestimates χ when externally mixed intermediate-hygroscopicity components break its binary more-/less-hygroscopic assumption**. The useful output is the size-resolved error distribution, which the authors then apply as uncertainty bounds to long-term HTDMA records at urban, continental and coastal sites. Model-derived error bars on an observational retrieval — the validation is only as good as PartMC's own composition realism, which the paper does not independently test.

doi: 10.5194/acp-26-10965-2026

Magdaong et al. 2026 ISPRS Ann Photogramm Remote Sens Spat Inf Sci

Criteria and an algorithm for siting a low-cost IoT air-quality network near roads

GIS multi-criteria analysis combining street aspect ratio, traffic emissions, global horizontal irradiance and road proximity into a 0–6 suitability surface for Quezon City, then a heuristic selecting nodes by joint suitability and population raster under a minimum-spacing constraint. A 40-sensor configuration covers **1.27–1.35 million residents, 23.0–24.4% of the city**, stably across weighting schemes while keeping spatial dispersion balanced. Limitation: coverage is defined geometrically against a population raster with no validation against measured concentration fields, so “covered” means “near a node”, not “well estimated”. Still the clearest statement this watch has seen that siting is an optimisation problem with a stated objective, and the criteria transfer directly to any near-road deployment in a low- or middle-income city.

doi: 10.5194/isprs-annals-x-5-w4-2025-315-2026

Masoom et al. 2026 Atmos Chem Phys

Long-range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features

Combined column remote sensing and in situ measurement of short- and long-range plumes across North America and Europe through the April–October 2023 fire season. The late-season episode is the payload: **concave AOD spectral curvature with the maximum above the 340 nm minimum wavelength, giving negative Ångström exponents below 500 nm**; accumulation-mode volume median diameters near 800 nm; and non-monotonic single-scattering albedo, rising through the UV–visible then flattening or falling at longer wavelengths, consistently across most stations. Limitation: a rare ~one-week event characterised observationally, with no closure between the retrieved optics and a measured composition, so the cause of the curvature is inferred rather than demonstrated. For distributed optical sensing the implication is that smoke-episode response cannot be extrapolated from background calibration.

doi: 10.5194/acp-26-10801-2026

Nimmala et al. 2026 Environ Monit Assess

Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling

Release–recapture with 200 GE switchgrass plants expressing orange fluorescent protein under a constitutive (PvUBI1) and a pollen-specific (Zm13) promoter, sampled to **1 km** by fixed and drone-borne collectors, with Lagrangian stochastic simulation driven by high-resolution wind. Emission rate estimated by combining simulated and measured concentrations; a strong diurnal trend correlates positively with wind speed, temperature and vapour pressure deficit and negatively with RH. Incorporating changing wind direction improved estimates under low-wind meandering — the transferable result for anyone inverting a dispersion model against sparse fixed sensors. Held back from the 2 August issue on its per-record [EDAT]; it is not a gap.

doi: 10.1007/s10661-026-15632-3

Orr et al. 2026 (JMIR) JMIR Pediatr Parent

In-home digital air quality monitors in families of children with asthma: feasibility and acceptability

Pilot mixed-methods study, **N = 20** families, combining interviews and surveys to assess whether a consumer in-home air quality monitor can be implemented in paediatric asthma care. Use was reported as both feasible and acceptable, with **affordability identified as the principal barrier to clinical adoption**. The report is very short and carries no exposure data, no adherence metric and no asthma outcome, so it establishes only that the deployment step is not the bottleneck. That is still the piece of the low-cost-sensing translation chain that is least often evidenced — most LCS literature stops at accuracy and never asks whether a clinical population will host the device. Needs a powered trial with objective device-uptime data before it can bear weight.

doi: 10.2196/92070

Oybek et al. 2026 Am J Appl Sci Technol

Chamber bounds on the temperature and humidity penalty, by sensor chemistry

Controlled-chamber factorial over six temperature (15–35 °C) and relative humidity (30–90%) combinations for an optical PM_{2.5} counter, a metal-oxide NO₂ sensor and an electrochemical CO₂ sensor. PM_{2.5} error rises from **1.2 µg/m³ at 30% RH to ~9 µg/m³ at 85% RH**; a simple linear correction recovers up to 74% of it. **Limitations are substantial**: one unit per chemistry, no replication, an unstated test aerosol, a venue with no visible peer-review record, and a linear correction applied to a process (hygroscopic growth) that is not linear. Carried as a magnitude anchor only — the number is consistent with better studies and useful for sizing an error budget, but nothing here should be cited as a correction method.

doi: 10.37547/ajast/volume06issue05-25

Qian et al. 2026 Atmos Meas Tech

AutoML calibration staged through an intermediate drift-reference sensor

Multi-stage indoor calibration framework linking deployed low-cost sensors to intermediate drift-correction reference sensors and finally to a reference-grade instrument, with separate models fitted for clean and pollution-event concentration ranges. Evaluated in a controlled chamber on two sensor models under diverse indoor sources and uncontrolled ambient conditions: $R^2 > 0.90$, NRMSE and sMAPE roughly halved against uncalibrated data, bias effectively removed. **Limitation: chamber conditions and an indoor concentration range**; the two-regime split is fitted, not physically motivated, and nothing establishes that the intermediate-reference tier is worth its cost against direct periodic co-location. The tiered-reference architecture is nonetheless the interesting part — it is a plausible answer to maintaining a large network with few reference-grade instruments.

doi: 10.5194/amt-19-603-2026

Rehbein et al. 2026 Clean Air J

Source identification and triangulation from a low-cost sensor network in Ile-Ife

Reanalysis of a published Nigerian LCS dataset using *k*-means over particle-number concentration profiles, resolving **six clusters** that separate day from night and local from regional influence. Stagnant conditions near industrial activity gave PM roughly 10× the daily mean (PM_{2.5} 30.6±40.7, PM₁₀ 83.4±259.2 µg m⁻³); cleanest conditions were daytime south-westerlies (PM_{2.5} 11.5±4.7), still above WHO annual guidance. Standard deviations exceeding the means signal a heavily skewed distribution that *k*-means handles poorly — the triangulation is indicative, not quantitative apportionment.

doi: 10.17159/caj/2026/36/1.24186

Reis 2026 Sensors

A low-power node that does its own inference and stops talking so much

Design and system-level optimisation of an embedded node integrating PM, gas, temperature and humidity sensing on a low-power microcontroller with a neural inference accelerator, using adaptive duty-cycling, hierarchical power domains and edge-level data reduction. On-device quantised fixed-point models perform three-class normal/anomalous/critical classification. Reported: **94% inference accuracy, 0.87 ms latency, ~2.9 mWh average power**, and an adaptive LoRaWAN strategy cutting transmissions by $\approx 88\%$ versus periodic reporting, enabling solar-battery autonomy. **Critical limitation: every number comes from MATLAB/Simulink simulation**, not a deployed node — so the accuracy figure describes a synthetic label distribution and the power figure omits the real cost of sensor warm-up, radio retries and thermal effects. Read as an architecture proposal, not a measurement.

doi: 10.3390/s26020703

Rubio et al. 2026 IEEE Consum Electron Mag

LoRa-connected micromobility pollution monitor for the Internet of Vehicles

On-board PM_x/CO₂/meteorology unit for personal mobility vehicles, evaluated across Cartagena's three-gateway LoRa network. The engineering contribution is compact low-power design plus LoRa in a genuinely mobile setting, where duty-cycle limits and Doppler normally break LPWAN links. No reference co-location and no uncertainty quantification are reported, so the pollution data quality is unestablished — the communications and positioning results are the defensible part.

doi: 10.1109/mce.2026.3684542

Samburova et al. 2026 Atmos Pollut Res

Effects of western US wildfire smoke on urban air quality in northern Nevada

METADATA ONLY — no abstract available in Crossref, Europe PMC or Semantic Scholar. Long-range transported wildfire smoke over an urban receptor is the same problem class as Jin et al. (ACP, 7 August) on aged smoke in Missoula; worth a re-check once the publisher deposits the abstract.

doi: 10.1016/j.apr.2026.103168

Sanders et al. 2026 Aerosol Sci Technol

Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles

Crossref deposit of 7 August, **metadata only — no abstract deposited anywhere in the retry chain**, so no summary is offered. Recorded here because surfactant control of hygroscopic growth is the physical mechanism underlying humidity correction in light-scattering PM sensors, and the supermicron size range is where nephelometric response is most sensitive to growth factor.

doi: 10.1080/02786826.2026.2703042

Silva et al. 2026 J Air Waste Manag Assoc

Accuracy, applications and limitations of low-cost sensors for indoor air quality

PRISMA 2020 review of PubMed, Scopus and ScienceDirect to March 2025; **24 of 2,736 screened studies** met inclusion and quality criteria, appraised with the Critical Review Form for Quantitative Studies. Optical particle counters and NDIR dominate; correlations against reference instruments are strong under controlled conditions and degrade with humidity, temperature and source type. Named unaddressed confounders: sensor aging and environmental cross-sensitivity. Conclusion for the programme — most devices do not currently meet regulatory compliance requirements, and standardised validation protocols do not exist.

doi: 10.1080/10962247.2026.2624795

Siu et al. 2026 Environ Res Health

Disparities in fine particulate matter (PM_{2.5}) monitoring capacity in eastern Canada

Locally adaptive kernel density estimation of total PM_{2.5} monitor density — regulatory plus low-cost — across Ontario, Quebec and Atlantic Canada, 2020–2024, with each local bandwidth set from the surrounding variability of surface PM_{2.5} and its sensitivity to resolving an exceedance. Mixed-effects logistic regression gives **25% higher odds of monitoring deprivation for residential instability and 54% higher odds for situationally vulnerable populations**, after adjusting for surface PM_{2.5} and emission-facility density; suburban Montreal and Toronto, the Windsor and Niagara border regions, southeast Kitchener and remote northwest Ontario are named. Limitation: monitor density is a proxy for information yield, and low-cost and regulatory devices are pooled without weighting by data quality, so a dense but uncalibrated cluster counts as capacity. The adaptive-bandwidth construction is directly reusable for designing where to add nodes rather than merely auditing where they are.

doi: 10.1088/2752-5309/ae87c0

Somnugpong 2026 Environ Monit Assess

Similarity-weighted seasonal-anchor Stineman interpolation for gappy air-quality series

SW-SAST keeps a local Stineman anchor and selectively borrows cross-day information when nearby daily analogues are compatible with the local estimate. Evaluated on UCI Air Quality and Beijing PM_{2.5} under MCAR, MAR and MNAR at 10–50% missingness, 30 repeated runs per condition, against a univariate LSTM baseline and with contiguous gap-length experiments. **It improves on plain Stineman for UCI, chiefly under MCAR and MAR, and fails to beat it on Beijing.** The contribution is explicitly a boundary condition on when seasonal borrowing helps, not a replacement interpolator — which is the right claim and an unusually disciplined one. Directly applicable to reconstructing low-cost-sensor series where dropouts are neither random nor uniform.

doi: 10.1007/s10661-026-15770-8

Suryawanshi et al. 2026 Int J Environ Res

Spatiotemporal variation and hotspot identification of PM_{2.5} from low-cost sensors

Abstract not indexed at the time of harvest; carried as a metadata-only record so the register is complete and the next rollup does not re-surface it as new. No claims are made about its content. Full text required.

doi: 10.1007/s41742-026-01169-3

Tian et al. 2026 Atmos Meas Tech

A heterogeneity index for when lidar-retrieved CCN vertical profiles stop being trustworthy

ARM ground-based Raman lidar plus surface CCN measurements retrieve vertically resolved CCN number concentration on the assumption of vertically homogeneous composition. A *k*-nearest-neighbours classifier on lidar ratio and linear depolarisation ratio infers dominant aerosol type by altitude, and a derived heterogeneity index quantifies vertical complexity. Validated against HI-SCALE aircraft in situ data: **retrieval closure degrades monotonically as the index rises**, so the metric functions as a self-diagnostic flag on the retrieval. Single campaign, so the index thresholds are not yet portable.

doi: 10.5194/amt-19-5051-2026

Tsai et al. 2026 IEEE Sens J

Vision-based vertical PM_{2.5} retrieval from a 5G-connected UAV

Dark-channel-prior atmospheric transmittance extracted from UAV imagery, fused with meteorology into a random-forest regression with 7-D polynomial feature engineering hosted on a 5G-core virtual host; Gemini used for semantic interpretation of the vertical profile. Reported $R^2 = 0.954$, RMSE $1.27 \mu\text{g m}^{-3}$. Treat with caution: no held-out spatial or temporal validation is described, the DCP proxy is illumination-dependent, and the generative-AI layer contributes interpretation, not measurement. Carried for the vertical-profiling concept, not the reported accuracy.

doi: 10.1109/jsen.2026.3695609

Wang et al. 2026 Environ Monit Assess

Traffic-induced environmental health assessment at intersections using roadside LiDAR

Roadside LiDAR extracts high-resolution vehicle trajectories at one urban and one suburban intersection, from which traffic emissions, noise and ground vibration are estimated and combined into a fuzzy comprehensive assessment score, with weights set by quadratic programming under ranking constraints, variable-weight grey entropy, and game theory to reconcile the two. The urban intersection scored **69.45 by day, falling to 57.12 at night (~18%)**, while the suburban site was stable at 72.80/74.54; population-dimension scores were consistently below environmental-dimension scores. Limitation: PM is *inferred* from trajectories through an emission model and never measured, so this is a modelled exposure surrogate validated against nothing — the elaborate weighting machinery sits on top of an unvalidated emissions estimate. Included because trajectory-derived emission proxies are an increasingly common input to near-road sensor fusion, and their weak point should be explicit.

doi: 10.1007/s10661-026-15750-y

Yan 2026 Sensor Review

What 90+ ML calibration studies actually agree on

Structured review of more than 90 peer-reviewed studies (2013–early 2024) across optical, electrochemical, metal-oxide and NDIR sensors and across PM, O₃ and NO₂. Consensus findings: ML calibration cuts error by >50% and lifts R^2 above 0.8–0.9; **optical PM sensors post-calibration cluster at R^2 0.8–0.95 with slopes near unity under long-term co-location**, while electrochemical NO₂/O₃ sit at 0.6–0.9 with much larger site-to-site spread. Neural networks and gradient boosting beat simple regression when data are plentiful; regression is more stable when they are not. Persistent problems named: drift, absent standard protocols, poor cross-site generalisation, and regulator resistance to black-box models. **Limitation: the search ends in early 2024** and the review is narrative in its weighting, with no quantitative pooling. Useful as a citation anchor for the claim that transferability, not fit, is the open problem.

doi: 10.1108/sr-10-2025-0761

Yang et al. 2026 Atmos Chem Phys

Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air

Ten years of observations from Penlee Point Atmospheric Observatory on the northeast Atlantic coast, from which **428 diurnal sea-breeze events** were identified, most frequent in spring and summer under low wind and strong irradiance. Daytime surface O₃, NO_x, CH₄, CO₂, SO₂, total aerosol number, PM_{2.5} and PM₁₀ were all elevated during events, with mean daytime O_x near **45 ppb** and increased regulatory exceedance for O₃. Accumulated ship-emitted SO₂ appears to drive new particle formation and a pronounced diurnal cycle in aerosol number. The monitoring payload: **excluding sea-breeze events reduces the diurnal amplitude and alters the seasonal cycle of O₃ and CH₄** under southwesterly flow — a coastal “background” site is conditionally not background. Limitation: one site, so the event classification and its consequences are not shown to transfer. Number-without-mass is precisely the regime a mass-based low-cost network cannot see.

doi: 10.5194/acp-26-10909-2026

Yaqoob et al. 2026 IEEE Access

Calibration that survives a change of city and a change of reference instrument

PurpleAir and Alphasense OPC-N3 in Antwerp, Oslo and Zagreb, hourly PM_{2.5} and PM₁₀, calibrated with Random Forest, a compact feedforward network, and HybriCal-AQ (a hybrid NN taking a standardised out-of-fold RF hint). Models are evaluated separately against TEOM, FIDAS and generic reference grades to mirror real reference availability, with a train-first split/impute/encode pipeline to avoid leakage. HybriCal-AQ wins on PM_{2.5} (Oslo-FIDAS/PurpleAir RMSE 1.79 → 1.18, R^2 0.94 → 0.97) while **plain RF usually remains better for PM₁₀** (Antwerp 4.76 vs 6.58). A sparse-reference test withholding 25% of reference observations probes robustness. **Limitation: three cities is not a transferability test** — every model is still trained and evaluated within-city, so this bounds instrument sensitivity, not spatial generalisation. The explicit reference-instrument stratification is the transferable methodological idea.

doi: 10.1109/access.2026.3656120

Yee et al. 2026 Atmos Meas Tech

UCB-GLOBES: open mass-spectral library for identified and unidentified organic aerosol

~27,000 GC/EI-MS spectra of trimethylsilyl derivatives of semi-volatile organic aerosol from US and Central Amazon ambient samples plus laboratory SOA, catalogued *without regard to known chemical identity* and annotated with sample provenance and predicted properties. Reanalysis of prior ambient data gave an 11% net increase in MS-based source categorisation for the southeastern US and lifted Central Amazon matches from 136 to 375 of 1,513 analytes. **~70% of UCB-GLOBES spectra match nothing** in the library or in NIST — the honest headline is the size of the unidentified fraction, not the size of the library.

doi: 10.5194/amt-19-5027-2026

Yukhymchuk et al. 2026 Aerosol Air Qual Res

Low-cost particulate matter sensor performance under polluted conditions in Northeast China

Crossref deposit of 7 August with **no abstract in Crossref, Europe PMC or Semantic Scholar**. Carried on title, journal and DOI only; **no summary is offered because none can be written without inventing content**. On title alone this is the single most directly relevant record of the window — LCS performance characterisation at the high concentrations where light-scattering response is least well constrained — and it should be revisited as soon as an abstract appears.

doi: 10.1007/s44408-026-00159-6

Zhao et al. 2026 Biomed Environ Sci

WHA air-LSTM: composite morbidity-driven pollution index for next-day outpatient forecasting

A five-level composite index is built *per city* from city-specific exposure–response relationships across >223.7 million hospital visits, then fed with temperature, humidity and visit history into an LSTM predicting next-day respiratory outpatient load. Beijing $R^2 = 0.963$, RMSE 53.5; removing the index degrades RMSE by 44.1%; performance holds on held-out cities. The index is fitted on morbidity and then used to predict morbidity, so the incremental-value claim is partly circular — the external validation is what carries it. Attractive template for turning a dense sensor network into an operational early-warning product.

doi: 10.3967/bes2026.062

Zhivkov et al. 2026 Atmosphere*How fast does a calibration decay with distance from its reference station?*

Twenty-four months of hourly data (Aug 2023–Jul 2025) from Sofia: five official reference stations and 22 AirThings low-cost units, four co-located. Random Forest calibration achieved R^2 0.53–0.75 across $PM_{2.5}$, PM_{10} , NO_2 and O_3 , improving on multiple linear regression by 40% (O_3) to 408% ($PM_{2.5}$). Leave-one-station-out spatial cross-validation yields pollutant-specific uncertainty growth rates α of **3.84–5.62% per km** (significant for PM_{10} and O_3 only, $p < 0.05$), applied to 18 non-co-located sensors to generate 1.2 M calibrated hourly values with 95% prediction intervals. **Limitation:** α is estimated from five stations in one city, so its significance for $PM_{2.5}$ is not established and the linear-in-distance form is an assumption. Still the most directly usable network design result in the window: 6 km anchor spacing keeps the midpoint uncertainty increase below 30%.

doi: 10.3390/atmos17040335

Exposure assessment & modelling**32 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (32 OF 32 RECORDS IN THIS SUBTOPIC)

Baldo et al. 2026 Atmos Chem Phys*Chemical and mineralogical composition of Southern African mineral dust*

First comprehensive characterisation of dust from a major and under-sampled source region. Laboratory-generated aerosol from soils at natural and emerging anthropogenic source areas, characterised by Si/Al, (Ca+Mg)/Al and K/Al ratios plus clay content. Those ratios separate arid western coastal sources from humid eastern inland sources and encode source-area weathering regime. The K-feldspar finding matters beyond composition — K-feldspar is the dominant mineral ice-nucleating agent, which links this record directly to the PINE/PIA instrumentation above. Laboratory-generated, so size distribution and mineralogy need not match what is actually emitted and transported.

doi: 10.5194/acp-26-10727-2026

Bhardwaj & Subramanian 2026 arXiv (preprint)*Identifiability-aware source apportionment for sparse city-scale sensor networks*

Reframes inventory-constrained apportionment as a wind-conditioned lagged inverse problem: time-varying source activity is represented on a low-dimensional nonnegative temporal basis, each source-basis coefficient produces a sensor-time fingerprint, and a separate low-dimensional background space is projected out. **Exact identifiability requires full column rank of the projected lagged response matrix**; noise-robust attribution is governed by its singular values, coefficient visibility, background absorption, pairwise coherence and ray distance. The practical payoff is diagnostic — you can test before deployment whether a proposed sensor geometry can separate two inventory groups at all. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.00050

Blanco-Villafuerte et al. 2026 (backfill) Curr Environ Health Rep*Hybrid low-cost-sensor, satellite and chemical-transport modelling of ambient $PM_{2.5}$ in Peru*

Review of national $PM_{2.5}$ exposure-model construction in LMICs, anchored on the NIH-funded Peru GeoHealth Hub's **176-monitor PurpleAir network** — 62.5% urban, 37.5% rural, across all 24 regions, established with national environmental agencies. The hybrid model fuses LCS, satellite retrievals, chemical transport output and machine learning; initial application to Lima reached $R^2 = 0.88$ against regulatory monitors for 2010–2023, with extension to daily national 5 km² fields for 2024–2026 in progress. R^2 against regulatory monitors in a single city is a weak validation for a national product, and the review is written by the network's own investigators. The deployment-ratio and institutional-integration detail is the transferable content. Crossref created 5 January 2026 — backfill, exclude from trend lines.

doi: 10.1007/s40572-025-00508-4

Buhendwa et al. 2026 Environ Pollut*Modelling the coupled $PM_{2.5}$ –ozone system in China: mechanisms, model biases, perspectives*

Review of the compound-pollution era in which $PM_{2.5}$ and O_3 share a precursor pool (VOCs, NO_x , SO_2 , NH_3) and single-pollutant control can worsen the co-pollutant — the “pollution seesaw”. The useful content is the bias inventory: chemical transport models **underestimate secondary organic aerosol by about 40%**, with substantial spread across models, seasons and regions; daytime HONO sources and OH reactivity are missing or understated; ammonia and episodic agricultural-burning inventories are uncertain; nocturnal boundary-layer mixing is poorly represented. These errors propagate into data assimilation and into machine-learning models trained on CTM output. Limitation: a narrative synthesis with no quantitative pooling, so the 40% figure is a reported central tendency, not a meta-analytic estimate. Directly relevant if CTM fields are used as priors or features in a sensor-fusion product.

doi: 10.1016/j.envpol.2026.128887

Cai et al. 2026 (ACP) Atmos Chem Phys*Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai*

Paired urban and suburban Shanghai sampling with high-resolution mass spectrometry, resolving **1,964, 1,914 and 2,689 organosulfur molecular formulas** across the sampled conditions and separating spatial, seasonal and day–night variation in formation pathway. Organosulfates are ubiquitous but their production regime is not: the comparison across contrasting sites is what isolates the pathways rather than the inventory itself. The limitation that matters for downstream use is that molecular formula assignment from exact mass does not fix structure, so pathway attribution is inferential. Relevant to sensing as a concrete demonstration that a large, chemically distinct fraction of urban $PM_{2.5}$ mass is invisible to any mass-only instrument.

doi: 10.5194/acp-26-11067-2026

Ding et al. 2026 Geo-spat Inf Sci*Urban–rural balanced estimation framework for $PM_{2.5}$ exposure across China*

Targets a structural bias rather than a model-fit problem: because ground stations cluster in cities, AOD-to- $PM_{2.5}$ retrievals systematically overestimate rural concentrations. Integrates AOD, reanalysis fields and ground measurements under data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples, then ensemble-averages across dynamically weighted background/observation settings. China 2015–2021: **clean-site RMSE 18.18 → 8.31 $\mu g m^{-3}$, national mean $PM_{2.5}$ down ~55%**, with corresponding correction of PM-attributable mortality in low-density areas. The 55% figure is the paper's own counterfactual, validated at clean sites that are still station locations — true rural accuracy remains unvalidated by construction. If even partly right, it implies an inherited upward exposure bias across the satellite-based exposure–response literature.

doi: 10.1080/10095020.2026.2624335

Helbich et al. 2026 Environ Int*Socioeconomic patterning of the urban external exposome depends on how the exposome is defined*

4,640 adults across Bulgaria's five largest cities, with residential NO₂, PM, traffic noise, green space, blue-space distance, impervious surface, walkability and night-time light. City-specific latent profile analysis was compared against composite indices, then both were regressed on individual- and area-level SES. High-risk profiles bundle low green space, high air pollution, noise and imperviousness *with* higher walkability. **Higher area-level SES was positively associated with high-risk profiles in most cities** — the inverse of the standard environmental-justice expectation — and composite indices only partially reproduced the latent-profile findings. The methodological warning is the finding: operationalisation choice changes the sign of the inequality conclusion.

doi: 10.1016/j.envint.2026.110419

Hu et al. 2026 Geo-spat Inf Sci*Fusion of ground monitoring, satellite retrieval and CTM for seamless hourly 1-km PM_{2.5}*

Fills satellite AOD gaps with continuous WRF-Chem AOD, injects physics-informed features (PM_{2.5} temporal components, AOD vertical structure) from the CTM, then predicts with a geospatially aware LightGBM that exploits monitoring-site spatial structure. Site-based cross-validation R^2 0.87, RMSE 13.42 $\mu\text{g m}^{-3}$, MAE 8.68 $\mu\text{g m}^{-3}$, with day-and-night coverage demonstrated on one pollution episode. Site-based CV overstates performance away from stations — the standard weakness — and an RMSE of 13.4 is large relative to the WHO annual guideline, so this is useful for episode mapping and weak for long-term exposure assignment. The physics-feature injection is the transferable step.

doi: 10.1080/10095020.2026.2680855

Kim et al. 2026 Environ Pollut*Wildfire plumes can lower downwind sulfate by scavenging OH*

CMAQ driven by the high-resolution FINN fire inventory, applied to the July 2014 Siberian wildfire episode. Long-range transport generally raised particulate matter across Northeast Asia, but during **26–29 July an anomalous sulfate decrease of up to $-1.56 \mu\text{g m}^{-3}$** appeared over Northeast China, the Shandong Peninsula and the Yellow Sea. The mechanism is competitive: fire-emitted CO and VOCs scavenge hydroxyl radicals, suppressing SO₂ oxidation. Source apportionment attributes 92.30% of the suppressed sulfate to inland anthropogenic SO₂ and 7.70% to oceanic sources. **Limitations: a single episode, model-only, with the mechanism inferred from the chemistry solver rather than measured OH**; FINN emission uncertainty propagates directly into the sign of the result. Relevant because it breaks the assumption that fire contributions to PM are additive — a source-apportionment scheme that assumes additivity will misattribute here.

doi: 10.1016/j.envpol.2026.128872

Li et al. 2026 Environ Monit Assess*Multi-media, multi-pathway PAH exposure and carcinogenic risk under three scenarios*

PAH concentrations in air, water, soil, dust and food compiled from Chinese studies published 2010–2024; average daily doses computed for diet, inhalation, ingestion and dermal contact; risk expressed as benzo[a]pyrene equivalents and incremental lifetime cancer risk, with Monte Carlo and sensitivity analysis. Industrial areas dominate air and water, urban areas dust and soil. **Dietary intake dominates both total exposure and carcinogenic risk**, and the industrial–urban–rural ranking is stable across dietary percentile scenarios even though absolute ILCR is not. Compiled from heterogeneous published measurements spanning 14 years — exposure misclassification is structural, not incidental.

doi: 10.1007/s10661-026-15753-9

Lu et al. 2026 Atmos Environ*Moving-vehicle wakes and noise barriers in traffic emission dispersion*

Metadata-only record. Surfaced by the Crossref by-journal leg as a new *Atmospheric Environment* deposit dated 2 August 2026; Elsevier deposits no abstract to Crossref and the article is not yet in PubMed or Europe PMC, so **no summary is offered and none should be inferred from the title**. Listed so that the window's coverage is complete and so the record can be picked up and summarised when an abstract becomes available. Authors: Lu, Li, Wang, Wu, Liu, Yang, Zhao, Zheng. The topic — vehicle-induced turbulence interacting with barrier geometry — is squarely relevant to near-road sensor siting, which is why it is carried rather than dropped.

doi: 10.1016/j.atmosenv.2026.122256

Madalipay et al. 2026 Atmos Pollut Res*Establishing PM_{2.5} conversion factors for South Korea at provincial level*

METADATA ONLY. Provincial PM_{2.5} conversion factors bear directly on how a sparse regulatory network is extrapolated to unmonitored areas, which is the same inference a low-cost network is deployed to replace. Priority re-check alongside the Madrid classroom study.

doi: 10.1016/j.apr.2026.103171

McCerery et al. 2026 Atmos Environ*Arctic transport of North American and Canadian wildfire aerosols*

METADATA ONLY. The only record in today's corpus with a polar or remote-marine receptor; long-range transport to a low-background site is where instrument detection limits, not calibration slope, become the binding constraint.

doi: 10.1016/j.atmosenv.2026.122289

Meena et al. 2026 Environ Pollut*Transplanted lichen trace-metal accumulation along a Jaipur urban gradient*

Phaeophyscia hispidula thalli collected from a wildlife-sanctuary reference site and transplanted to ten urban and one reference location in Jaipur for 90 days over the March–June 2024 pre-monsoon, with Zn, Pb, Cd, Cr, Cu and Ni by ICP-MS and chlorophyll and carotenoid concentrations by spectrophotometry. Urban sites accumulated more metal, and **accumulation tracked chlorophyll loss while carotenoids rose** at several exposed sites, consistent with a photoprotective adjustment. PCA and hierarchical clustering resolved a spatial gradient. **No co-located instrumental PM measurement means the biomonitor is never calibrated against a concentration**, so this yields a relative ranking of sites, not an exposure estimate. Useful as an ultra-low-cost spatial screening layer where no network exists; not a substitute for one.

doi: 10.1016/j.envpol.2026.128919

Mushore et al. 2026 Atmos Environ X*Daily outdoor PM_{2.5} and the road-edge effect across The Gambia, Kenya and Mozambique*

Multi-temporal personal exposure measurements (2022–23) interpolated to continuous surfaces by inverse-distance weighting, validated leave-one-out at $R^2 > 0.55$, MAE $< 4 \mu\text{g m}^{-3}$, RMSE $< 10 \mu\text{g m}^{-3}$; multi-distance buffers at 10 m intervals to 1000 m from major roads. Most sites exceed WHO guidance, The Gambia highest across all land-cover classes. **The road-edge gradient reverses sign between sites** — PM_{2.5} falls with distance in Rabai and Mozambique and *rises* elsewhere, implicating wetland, fire and industrial sources that a traffic-proximity exposure proxy would misattribute entirely. IDW over sparse personal measurements is a weak spatial model; the buffer result is the transferable part.

doi: 10.1016/j.aesoa.2026.100463

Myat et al. 2026 Environ Pollut

Ground, satellite and model assessment of PM_{2.5} and attributable mortality in Myanmar

Multi-source assessment for a least-developed country with minimal regulatory monitoring: two Yangon ground stations 2019–2024, TROPOMI HCHO and NO₂ nationally, and a generalised additive model for meteorological dependence. Mean PM_{2.5} was **24.4 µg/m³ (SD 18.6)**, rising to **39.4 µg/m³** in the November–February winter, with sulfate the leading measured component year-round and nitrate and Ca²⁺ seasonally. Compliance with the 5 µg/m³ WHO guideline would have averted **2,106 (95% CI 345–3,818) premature deaths annually** among adults ≥30 in Yangon over 2020–2021. TROPOMI HCHO/NO₂ ratios indicate NO_x-sensitive ozone chemistry nationally. **Two stations cannot represent a city of five million**, and the health impact assessment inherits every assumption of the imported concentration–response function.

doi: 10.1016/j.envpol.2026.128918

Nguyen et al. 2026 Eng Res Express

Satellite-driven PM_{2.5} prediction: machine learning and explainable AI

Systematic review of 2015–2025 work integrating satellite products (LEO and geostationary AOD, top-of-atmosphere reflectance, Ångström exponent) with ML/DL. Finds Random Forest still a robust benchmark, hybrid and transformer architectures better at spatiotemporal dependence, and AOD gap-filling plus normalisation the dominant determinants of model stability; positions XAI as the route to physical plausibility. No pooled quantitative comparison and no protocol registration, so the ranking claims are narrative. Adds little beyond Hu and Ding above; retained as a citation anchor for the physics-guided-learning argument.

doi: 10.1088/2631-8695/ae7028

Parajuli et al. 2026 Atmos Environ

Which soil NO_x parameterisation you pick changes modelled ozone and PM

As fossil-fuel NO_x declines, ozone becomes more sensitive to non-fossil nitrogen, yet most air-quality models still use simple soil-NO parameterisations. This study compares Yienger & Levy and BDSNP against the mechanistic FEST-C* fertiliser emissions tool, propagating each through CMAQ over the United States and evaluating against rural NO₂ monitors and satellite NO₂ columns. FEST-C* agricultural soil NO/HONO was **double YL and two-thirds of BDSNP**, improved agreement at some rural sites, and had intermediate correlation but the lowest bias over agricultural areas. **Limitation: the PM response is reported only qualitatively** — the paper is NO_x- and ozone-centred, so a PM-focused reader gets the direction and not the magnitude. Carried because it identifies a structural uncertainty in a model layer that PM exposure surfaces routinely inherit without examination.

doi: 10.1016/j.atmosenv.2026.122225

Pavlovic et al. 2026 J Air Waste Manag Assoc

Aligning satellite Earth-observation capability, use case and capacity for air-quality health

Twenty-three-author synthesis from the GEO Health Community of Practice Air Quality and Respiratory Health Work Group. Argues satellite EO translates into public health benefit only where three things align: technical capability of the retrieval, a defined use case, and the human, computational, operational and governance capacity to sustain a solution. Surveys applications from long-term trend tracking and epidemiological exposure assignment to short-term emissions detection and early warning, then names gaps in each dimension. A consensus document rather than new evidence — useful as a citation anchor for why a given satellite product is or is not fit for a stated purpose.

doi: 10.1080/10962247.2026.2698602

Phalen et al. 2026 Inhal Toxicol

Inhaled aerosol dosimetry update

Synthesis of the Fourth International Aerosol Dosimetry Conference (Irvine, 2024) covering computational deposition models, in vitro and in vivo exposure systems, and deposition variability. Reported advances: integration of model types to span all major respiratory-tract regions; improved in vitro particle delivery and dose measurement; the upper airways recognised as a significant entry route. Two findings matter for ambient PM work — **dose variability emerged as the critical cross-cutting issue**, and **deposited ultrafine particles may acquire surface coatings after deposition and translocate into tissue, including fetal tissue**. Limitation: a conference report, not a systematic review — coverage reflects who presented, there are no effect sizes and no pooling, and the fetal-translocation claim is stated without the primary evidence. Useful mainly as a statement of where dosimetric uncertainty sits, which is the term that converts a measured ambient concentration into an internal dose.

doi: 10.1080/08958378.2026.2691193

Pinakana et al. 2026 NPJ Clean Air

Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools

Low-cost sensors deployed at schools in Roma and Rio Grande City, Starr County, on the US–Mexico border for **146 days, November 2023 to April 2024**, corrected by collocation-based calibration against a research-grade instrument and read alongside meteorology, back trajectories and satellite smoke and fire products. Daily mean PM_{2.5} was significantly higher on smoke-impacted days ($p < 0.001$), with **site-level increases of 8.61% to 54.9%** — a sixfold spread across sites within one small county, which is itself the finding: transboundary smoke impact is not spatially uniform at the scale a single regulatory monitor would represent. Trajectories attribute the source to Mexican wildfire emissions. Limitation: one season, no indoor measurement despite the school framing, and attribution rests on trajectory plus satellite coincidence rather than on composition. A clean template for low-cost sensor deployment in an under-monitored community.

doi: 10.1038/s44407-026-00096-z

Rai et al. 2026 Environ Monit Assess

Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993–2023

Multi-sensor remote sensing: Landsat LST and urban thermal field variance index for 1993, 2002, 2014 and 2023, with MODIS AOD and LST from 2002–2023 for the diurnal response. **Urban extent grew by 171.15 km² over 30 years** with sharp losses of fallow and vegetated land, and the thermal environment polarised — fallow land carries the worst thermal comfort, water and vegetation the best, and NDVI/NDBI track the expected cooling and warming signs. **The AOD–LST correlation is dual-signed**, which the paper reports but does not mechanistically decompose into aerosol direct radiative effect versus co-varying surface change; that confounding is the main threat to the inference. Read as a heat-planning input rather than a PM exposure product — AOD here is a columnar quantity never converted to surface concentration.

doi: 10.1007/s10661-026-15748-6

Raptaki et al. 2026 Atmos Environ

Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation

METADATA ONLY. Enters on the secondary-organic-aerosol precursor link rather than on any direct PM measurement; tier C.

doi: 10.1016/j.atmosenv.2026.122283

Roky et al. 2026 Environ Pollut*Distance-resolved lead dispersion and crop uptake from open battery-plate burning*

Controlled field burns of lead-acid battery plates in Kushtia, Bangladesh (three plates per event, 149.7 ± 4.5 g total mass loss), monitored on a triangular sampling grid to 45 m with simultaneous $\text{PM}_{2.5}/\text{PM}_{10}$, passive Pb deposition on filter papers, and Pb accumulation in red amaranth. All three response variables showed strong distance-dependent gradients with significant distance, week and interaction effects ($p < 0.0001$); **$\text{PM}_{2.5}$ tracked both deposition and crop Pb more strongly than PM_{10}** , and background plots stayed below detection, supporting atmospheric deposition as the dominant pathway. Crop-response model $R^2 = 0.948$. **Limitation: the threshold distance is not a single number** — the Codex leafy-vegetable benchmark is fresh-weight and the plant data dry-weight, so it had to be evaluated across a plausible moisture range, and 45 m does not bound the plume. A clean demonstration that a short-range $\text{PM}_{2.5}$ gradient is a usable proxy for a metal deposition field.

doi: 10.1016/j.envpol.2026.128853

Temu et al. 2026 Environ Sci Pollut Res*A six-domain framework for lichen biomonitoring including symbiont and microbiome response*

Narrative synthesis proposing that lichen biomonitoring be extended from the classical triad of pollutant accumulation, physiological injury and community composition to six domains, adding functional traits, symbiotic interaction and lichen-associated microbiome dynamics as **earlier-responding indicators** than bulk accumulation. The argument that microbiome shifts precede visible physiological injury is plausible and is the part worth testing, but the review supplies **no quantitative comparison of detection limits or response latency across the six domains**, so the central claim of earlier warning is asserted rather than demonstrated. The authors name geographic bias, species-specific response and absent methodological standardisation as open problems — the same standardisation gap that limits low-cost sensor networks, and for structurally similar reasons.

doi: 10.1007/s11356-026-38123-x

Wandji Nyamsi et al. 2026 Atmos Chem Phys*Broadband aerosol optical depth reconstructed from sunshine-duration records since 1900*

Uses sunshine duration as a proxy for broadband AOD together with cloud-cover observations from **2,700 sites**, producing the first historical BAOD estimates for several regions back to the late 19th century. The motivation is the persistent spread in modelled anthropogenic aerosol radiative forcing. The proxy is indirect and confounded by cloud regime and 120 years of instrument-network change; the authors' own framing — a constraint on century-scale trend, not a concentration product — is the right reading.

doi: 10.5194/acp-26-10767-2026

Wang et al. 2026 Atmos Chem Phys*Direct thermal versus emission-mediated pathways in heatwave-driven O_3 and SOA increases*

Explicitly decomposes the two routes by which heatwaves raise ozone and secondary organic aerosol: direct acceleration of photochemistry, versus temperature-driven biogenic VOC and soil-NO emissions. **Direct thermal and meteorological pathways account for over 80% (normalised) of the national increases**; ozone-temperature sensitivity peaks in the Yangtze River Delta ($0.66 \text{ ppb } ^\circ\text{C}^{-1}$) and Pearl River Delta (0.95). Model-based attribution, so the decomposition inherits the chemical mechanism's biases — but the direction of the result argues against the common assumption that BVOC enhancement is the dominant heatwave pathway.

doi: 10.5194/acp-26-10789-2026

Xu et al. 2026 Atmos Chem Phys*Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities*

Simultaneous winter measurement of $\text{PM}_{2.5}$ -bound nitroaromatics in 11 cities. Nitrophenols and nitrocatechols dominate, with city-varying relative abundance; totals are higher in northern cities, attributed to biomass and coal combustion. Secondary processes dominate formation everywhere, but the paper's actual target is the claim — built from chamber and single-site work — that relative humidity or aerosol liquid water *drives* that formation. At large-scale field level the relationship is conditioned by multi-factor interactions rather than being a clean RH dependence. A useful corrective to chamber-derived parameterisations.

doi: 10.5194/acp-26-10751-2026

Yadav et al. 2026 Research Square (preprint)*Airborne endotoxin on PM_{10} in Agra: an 8-fold seasonal swing that PM mass does not track*

Full calendar year (Jan–Dec 2022) of ambient endotoxin by chromogenic LAL assay with concurrent bacterial and fungal bioaerosol, PM_{10} and meteorology in the semi-arid Indo-Gangetic Basin. Monthly means run 0.92 ± 0.35 (June) to $7.35 \pm 1.48 \text{ EU m}^{-3}$ (December) on winter stagnation. Endotoxin correlates strongly with bacterial concentration ($r_s = 0.93$) and **not at all with PM_{10} mass** ($r_s = 0.10$, $p = 0.75$) — mineral dust is not the carrier here. USEPA-model hazard quotients stay well below unity (max 0.0198, children, December). Preprint, single city, LAL assays are notoriously matrix-sensitive; the decoupling from mass is the durable point.

doi: 10.21203/rs.3.rs-10523351/v1

Yu et al. 2026 Environ Pollut*Street-canyon increment to near-road NO_2 , resolved for a whole city*

Coupled WRF-CMAQ-ADMS framework with a purpose-built street-scale traffic emission inventory for Hong Kong, run in paired configurations with and without the street-canyon module to isolate the canyon-induced increment ΔC . Hotspots (segments with $\Delta\text{C} > 5 \mu\text{g}/\text{m}^3$, roughly a 1% increment in long-term NO_2 -attributable added risk) account for **12.2–18.8% of roads** in the four worst districts. Canyon effects are seasonal — mean hotspot ΔC rises **19–26% from summer to winter** under weaker ventilation — and about **184,000 residents** are exposed to substantial winter increments. **Limitations: NO_2 only, no PM increment reported**, and the canyon module is validated indirectly through the coupled system rather than against canyon-resolved measurements. The reason to read it here is siting: it identifies, at segment resolution, where a sparse regulatory network is structurally blind and where dense low-cost deployment would actually add information.

doi: 10.1016/j.envpol.2026.128871

Zhang et al. 2026 Environ Sci Technol *H_2O_2 -driven sulfate formation at air–water interfaces: stepwise mechanism and accelerated kinetics*

Theoretical study of the dominant global route to particulate sulfate. At the air–water interface the pre-reaction complex is favoured and the reaction proceeds stepwise — nucleophilic attack of HSO_3^- on H_2O_2 to a HOOSO_2^- intermediate, then isomerisation to sulfate — with a **rate-determining barrier of 4.1 kcal/mol**, lower than in bulk, and with the rate-determining step itself shifting. Interfacial electric field and partial solvation together account for **89% of the barrier reduction**. Limitation: entirely computational, with no experimental kinetic validation, and the idealised interface is not a deliquesced particle surface. It matters because rising H_2O_2 from wildfire activity makes this pathway more important, and because sulfate is both a large $\text{PM}_{2.5}$ mass fraction and one of its more hygroscopic — therefore optically most humidity-sensitive — components.

doi: 10.1021/acs.est.5c17902

Zheng et al. 2026 Atmos Environ*Spatially heterogeneous drivers of winter PM_{2.5} in China from synchronous observation and explainable ML*

Crossref deposit of 7 August, **metadata only** — Elsevier deposits no abstract to Crossref and the record is not yet in Europe PMC, PubMed or Semantic Scholar. No summary is offered. Logged because synchronous multi-site observation paired with explainable machine learning is the design that the sensing literature most often lacks: it is the combination that could distinguish genuine spatial heterogeneity in drivers from instrument heterogeneity across a network.

doi: 10.1016/j.atmosenv.2026.122285

Cardiovascular & metabolic**7 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Leiser et al. 2026 JACC Adv*Ambient PM_{2.5} and preclinical change in cardiac structure and function*

1,630 participants with echocardiography at two visits (2008–11, 2015–18), median 4.3 years apart; residential PM_{2.5} and NO₂ from spatio-temporal models as annual means preceding each visit, adjusted for baseline and time-varying covariates. Per IQR of PM_{2.5}: **global longitudinal strain +0.52% (0.20–0.83), LVEF –1.05%/year (–1.73 to –0.37), relative wall thickness +0.12 unit (0.05–0.19), left atrial volume index –1.39 (–2.25 to –0.54)**. NO₂ moves the same direction, smaller. The LAVI direction sits awkwardly with a concentric-remodelling reading and the paper does not reconcile it; treat the strain and RWT results as the load-bearing ones.

doi: 10.1016/j.jaccadv.2026.103104

Li et al. 2026 (Environ Int) Environ Int*PM_{2.5} constituents, hepatic fat and stiffness in youth, modified by PNPLA3 genotype*

113 Los Angeles Latino adolescents with obesity, MRI-measured hepatic fat fraction and liver stiffness, serum metabolomics and lipidomics, and visit-year mean exposure to **15 PM_{2.5} components** at residence; linear regression plus g-computation for mixture effects with PNPLA3 genotype as modifier, and moderated mediation. A simultaneous IQR increase across all components was associated with **10.0% higher hepatic fat fraction**, with elemental carbon, bromine, copper, potassium and lead as leading contributors — a combustion and traffic signature. **Effects concentrated in PNPLA3 GG carriers**, who showed 550 mediating metabolite features against 196, broader pathway enrichment, and the only significant total mediation. Replication in 81 mixed-ethnicity young adults held for EC and Br. Small discovery sample, cross-sectional, modelled rather than personal exposure — the genotype interaction is the hypothesis, not the estimate.

doi: 10.1016/j.envint.2026.110438

Liu et al. 2026 China CDC Wkly*Long-term ozone and cardiovascular disease: exposure windows, sex and region*

Meta-analysis of cohort studies from PubMed, Web of Science and Embase, stratified in the way most prior reviews were not. Per 10 µg/m³ ozone: CVD mortality RR 1.0105 (1.0027–1.0185), morbidity RR 1.0578 (1.0289–1.0874). The exposure-window split is the result: **warm-season ozone RR 1.0186 (1.0119–1.0253) versus an annual-average RR of 0.9971 (0.9781–1.0164)** — annual averaging destroys the signal. Female RR 1.1135 (1.0057–1.2328) versus male 1.0707 (0.9855–1.1633); Asia estimates are imprecise. Co-pollutant record; no PM estimate is reported. The averaging-window lesson transfers directly to PM network design.

doi: 10.46234/ccdcw2026.161

Sewor et al. 2026 Environ Res Health*Personal household-air-pollution exposure and blood pressure in rural Rwandan children*

622 children aged 8–15 from 626 biomass-cooking households, 48 h personal PM_{2.5} and black carbon by Ultrasonic Personal Aerosol Sampler, then standardised BP measurement. Median 48 h PM_{2.5} **192.9 µg/m³** (IQR 119.7–336.0), BC 9.85 µg/m³. **No association with SBP or DBP** (PM_{2.5}–SBP 0.51 mmHg per 211.7 µg/m³ IQR, 95% CI –0.46 to 1.49) and no clear effect modification by age or sex. Yet 24% of the children were hypertensive or had elevated BP against US reference percentiles. Cross-sectional, and the exposure range sits entirely above the region of the global exposure–response curve where slope is expected — a null that constrains where measurement effort is epidemiologically productive.

doi: 10.1088/2752-5309/ae8ff9

Shi et al. 2026 Catheter Cardiovasc Interv*Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates*

GBD 2023 descriptive burden analysis of ischemic heart disease incidence, mortality and DALYs across the G20, 1990–2023, with EAPC and joinpoint regression, population attributable fractions for modifiable risks, ARIMA and Bayesian age-period-cohort projection to 2050, and XGBoost with SHAP for DALY determinants. **8.52 M cases and 5.81 M deaths in 2023**; age-standardised incidence and mortality fell (EAPC –1.61 and –1.49) but incidence rose in women aged 15–49, with Russia highest on all three age-standardised rates and Japan projected to peak again around 2032. High systolic blood pressure dominated the risk attribution, with **particulate matter and lead named as the prominent environmental contributors**. Limitation: GBD population attributable fractions are modelled from the same exposure–response literature this watch tracks, so this is not independent evidence of a PM effect — it is that literature propagated through a burden model, and the machine-learning layer adds interpretability, not identification. Tier C: read as context, not as evidence.

doi: 10.1002/ccd.70783

Zhao et al. 2026 Medicine (Baltimore)*Ischaemic stroke burden and attributable risk factors in adults aged 20–54, 1990–2021*

GBD 2021 extraction for ischaemic stroke in the 20–54 age band, with age-period-cohort models and population-attributable fractions by risk factor. In 2021 there were **16,301,644 prevalent and 1,324,743 incident cases** (432.46 and 35.14 per 100,000), 105,859 deaths and 6.97 million DALYs. **Prevalence and incidence rose over the period while mortality and DALYs fell**, with burden concentrated in high-middle SDI regions, Eastern Europe, Central Asia and parts of North Africa and the Middle East. Air pollution enters only as one of the GBD attributable risk factors, so this contributes no exposure–response estimate and inherits every GBD modelling assumption, including sparse primary data in the highest-burden regions. Carried for the age-band signal: young-adult stroke burden is rising in exactly the regions where PM exposure is highest and monitoring is thinnest.

doi: 10.1097/MD.00000000000050104

Çulpan et al. 2026 Rev Environ Health*Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis*

PRISMA-guided update over prior reviews plus PubMed and Web of Science to 27 March 2025; 21 studies, 990,157 participants in the qualitative synthesis, with meta-analysis restricted to the fine and coarse fractions and NO₂ because cohort data for SO₂, O₃ and CO were too sparse.

Fine PM: +0.59 mmHg systolic per 10 µg/m³ (0.28–0.90), the only estimate consistent across sensitivity analyses. The coarse fraction gave nominally tighter systolic (+0.28, 0.25–0.31) and diastolic (+0.27, 0.24–0.30) associations that *did not survive* sensitivity analysis — an implausibly narrow interval for a pooled estimate, which usually indicates a single dominant study. NO₂ was null for both. Several pollutants raised the odds of hypertension as a dichotomous outcome. Two limitations are load-bearing: exposure increments and hypertension definitions were not harmonised across primary studies, and no multipollutant model could be fitted because the primary studies did not report exposure covariance, so the pollutant-specific attribution is weaker than the point estimates suggest. Note also that the PubMed deposit strips subscripts from every particle-size label, so fraction assignment here is inferred from the stated meta-analytic scope.

doi: 10.1515/reveh-2026-0032

Neuro / mental health**2 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (2 OF 2 RECORDS IN THIS SUBTOPIC)

Jahanshahi et al. 2026 Int J Popul Data Sci*Dementia onset and long-term PM_{2.5} in a genuinely low-pollution setting*

Northern Ireland census data linked to administrative prescription records, 2010–2016, with dementia onset proxied by first dispensed dementia medication — an endpoint that avoids diagnostic-coding variability at the cost of missing untreated cases. Cox proportional hazards models estimated the effect of five-year moving-average PM_{2.5} and NO₂ with extensive confounder adjustment. **Clear positive associations in both unadjusted and adjusted models**, persisting across age and sex subsamples and more tentative below age 70. **Limitations: the abstract reports no numerical hazard ratios or intervals**, so the effect size cannot be entered into the forest plot; prescription-based ascertainment under-counts and lags true onset; and exposure is residence-assigned with no mobility term. The contribution is contextual — it extends the dose–response into a low-concentration range where a threshold has repeatedly been proposed and not found.

doi: 10.23889/ijpds.v11i15.3644

Lin et al. 2026 Genes Dis*A lung–brain axis hypothesis for pollution-associated epilepsy*

Hypothesis-driven review proposing that inhaled air pollutants disrupt lung microbiome homeostasis and that this dysbiosis — by analogy with the gut–brain axis — contributes to epileptogenesis. Assembles three strands: environmental associations between air pollution and epilepsy susceptibility, links between lung-intrinsic microbiota dysregulation and neurological dysfunction, and candidate molecular, immunological and neural pathways by which the lung–brain axis could modulate seizure threshold. **This is a proposal, not evidence.** No new data; the pollution–epilepsy associations it rests on are sparse and confounded, and no step of the chain (PM → lung dysbiosis → neural excitability) has been demonstrated in the same system. Carried at tier C because the mechanistic-toxicology cluster has been empty this window and because it names testable intermediate endpoints — lung microbiome composition after controlled exposure is measurable now.

doi: 10.1016/j.gendis.2026.102240

Respiratory & allergic**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Cantuaria et al. 2026 Environ Health Perspect*Source-specific PM_{2.5}, UFP, EC and NO₂ and incident COPD in two pooled Danish cohorts*

159,769 participants from the Diet, Cancer and Health cohort and the Danish National Health Survey, with complete residential address histories driving 10-year time-weighted exposures split into local-traffic and other-source contributions; Cox models adjusted for demographic, socioeconomic and lifestyle factors including smoking. Per IQR of total exposure: PM_{2.5} **HR 1.11 (1.05–1.17)**, NO₂ 1.08 (1.04–1.13), UFP 1.05 (0.99–1.11), EC 1.02 (1.00–1.05). **The source split inverts across metrics** — non-traffic PM_{2.5} was the more harmful fraction while traffic-derived UFP and EC dominated theirs. Associations were stronger in women, never-smokers and prior asthma. Residual confounding by smoking intensity is the live threat given COPD's dominant cause. Directly relevant to siting: a traffic-gradient network resolves the wrong PM_{2.5} fraction and the right UFP one.

doi: 10.1021/EHP.6c00128

Haque et al. 2026 Health Sci Rep*PM and respiratory disease in Bangladesh: what 42 studies show*

Narrative review of 42 studies (2013–2025) drawn from seven databases on fine and ultrafine PM and gaseous pollutants and respiratory outcomes in Bangladesh. PM_{2.5} in Dhaka, Chattogram and Khulna is reported at **8–18× WHO guideline levels**, with persistent hotspots around brick kilns, industrial zones, transport corridors and seasonal biomass burning, and consistent associations with asthma, COPD, bronchitis, acute respiratory infection and hospital admission, concentrated in children, older adults and occupational groups. **Limitations: narrative, not systematic** — no risk-of-bias assessment, no pooling, and heavy reliance on primary studies with sparse monitoring infrastructure, so the exposure side of most included work is itself weak. Carried because it documents where the burden is largest and the measurement network is thinnest, which is the standing argument for low-cost network deployment in South Asia.

doi: 10.1002/hsr2.72925

Hur et al. 2026 Environ Pollut*Pollen modifies the SPM–mortality association across eight Kyushu cities*

Two-stage time-series over 262,288 all-cause deaths, eight cities, pollen seasons (Feb–Apr) 1989–2014, with an explicit SPM × pollen interaction term in stage one and random-effects pooling in stage two. Respiratory mortality at lag 1: **RR 1.010 (1.000–1.019)** overall, **1.021 (1.005–1.037)** on days above the 75th-percentile pollen concentration against **1.003 (0.987–1.018)** below, stable across alternative cutoffs. The threat to inference is exposure misclassification in the modifier, not the exposure: pollen is measured at a single station per city and the interaction is fitted on a binary split of a highly skewed distribution, so the contrast is a lower bound on a continuous effect. Suspended particulate matter is a Japanese regulatory metric with a 100% cut near 10 µm and is not interchangeable with PM_{2.5}. For distributed sensing the implication is concrete: an optical counter with a fluorescence or coarse-mode channel can resolve the modifier and the exposure on the same node, which no regulatory network in this study did.

doi: 10.1016/j.envpol.2026.128933

Hussein et al. 2026 J Glob Health

PM₁₀ and emergency asthma presentations in Petaling Jaya, Malaysia

Four years of ED records (Jan 2014 – Apr 2017) from a Klang Valley hospital against pollutant and meteorological data from nearby stations; generalised additive models plus distributed lag non-linear models for delayed and non-linear PM₁₀ effects. Asthma exacerbations were 2.3% of all ED visits. Relative to the Malaysian standard, daily mean PM₁₀ of 200 µg/m³ was associated with a 21.3% increase (95% CI 2.2–44.0) and 300 µg/m³ with 34.0% (4.7–71.7); the cumulative lag 0–7 exposure–response turns upward from about 150 µg/m³, and the lag structure shifts from days 2–3 at 100 µg/m³ to same-day at ≥200. Single hospital, single monitoring catchment, wide intervals — the authors correctly label the associations exploratory. The threshold shape is the transferable part: it argues for alert triggers well below current national standards.

doi: 10.7189/jogh.16.04249

Jiang et al. 2026 J Cent South Univ Med Sci

Short-term PM and respiratory-infection hospitalisation burden in kidney transplant recipients

Retrospective time-series over 874 kidney transplant recipients hospitalised for respiratory tract infection at the Third Xiangya Hospital, 2017–2023, with high-resolution daily exposure to six pollutants over the 7 days before symptom onset and DLNMs for lagged and cumulative associations. PM_{2.5} showed the strongest association with burden: at 160 µg/m³ cumulative exposure, hospitalisation cost rose by 9,544.5 yuan (95% CI 2,548.8–21,637.7); PM₁₀ showed a dose-dependent trend on length of stay (RR 1.63 at 300 µg/m³) whose interval, 0.18–15.94, is so wide it carries no information and is deliberately excluded from the forest plot. SO₂ above 75 µg/m³ also raised burden; CO, NO₂ and O₃ were null. Single centre, immunosuppressed population, ambient exposure assigned without residential history. Economic rather than clinical endpoints, which is unusual and useful.

doi: 10.11817/j.issn.1672-7347.2026.260089

Orr et al. 2026 (EHP) Environ Health Perspect

Wildfire-season PM_{2.5} and influenza / ILI risk across six western US states

Generalised linear distributed-lag models on state health-department influenza and ILI counts from Arizona, Colorado, Montana, Nevada, Oregon and Washington, 2010–2019, testing both preceding-wildfire-season exposure on the following flu season and short-term exposure within it. Per 10 µg/m³: CO 1.067 (1.056–1.078), AZ 1.061 (1.026–1.100), OR 1.049 (1.041–1.057), MT 1.038 (1.013–1.063) — all four states with laboratory-confirmed influenza data — against NV 1.005 (0.920–1.097) and WA 0.884 (0.842–0.919), the two with syndromic ILI only. The effect boundary coincides exactly with the outcome-ascertainment boundary, so the geographic reading is almost certainly the wrong one. Ecological design, no individual exposure, and wildfire-attributable PM_{2.5} is itself a modelled quantity.

doi: 10.1021/EHP.6c00326

Shukla et al. 2026 BMJ Open

Hourly NO₂ and emergency admissions for acute asthma, Birmingham and Solihull

Poisson generalised additive model with a distributed-lag non-linear term on hourly counts — 18,943 adult asthma admissions to four acute hospitals, June 2016–May 2022, linked through the PIONEER data hub. The mean 18 µg/m³ NO₂ exposure over lag 0–24 h carried RR 1.13 (1.01–1.26) for daytime admissions, with the association concentrated at a 3–6 hour lag and a steeper gradient in the most deprived quintile. Exposure is assigned from fixed monitors to residence, so the hourly resolution is nominal rather than personal — time-activity displacement over a 3–6 h window is precisely the scale at which residential assignment fails, which biases toward the null. The pollutant is NO₂, so this enters the watch on method rather than on substance: it is the cleanest recent demonstration that the health-relevant exposure signal lives below the 24-hour averaging period that filter-based reference methods impose.

doi: 10.1136/bmjopen-2025-110612

Solazzo et al. 2026 Hyg Environ Health Adv

Season and pollutant interact on the nasal microbiome

Longitudinal repeated-measures design: 26 healthy Italian adults sampled weekly for three weeks in winter and three in summer, with 16S rRNA sequencing (124 samples) and whole-genome shotgun sequencing (141 samples), against weekly indoor TSP and outdoor particulate matter, black carbon, benzene and CO. The microbiota was stable within season but differed between seasons, with winter enrichment of *Moraxella* — notably among women with children. Higher pollutant levels were negatively associated with commensal abundance, particularly *Corynebacterium*, and significant season × pollutant interactions appeared: in summer, *Staphylococcus epidermidis* and *Cutibacterium granulosum* were negatively associated with PM. Sex explained the most variance in diversity. Limitations: *n* = 26 with many taxa tested, exposure assigned from area monitors rather than personal sampling, and season is collinear with everything from heating to indoor occupancy. Hypothesis-generating; the interaction structure is the part worth chasing with a larger cohort and personal monitors.

doi: 10.1016/j.heha.2026.100189

Wang et al. 2026 Res Sq (preprint)

PM_{2.5} constituents and daily fever-clinic visits in Jinshan, Shanghai

Time-series generalized additive models linking 77,637 fever-clinic visits in 2024 to PM_{2.5} mass and speciated constituents, with single-day lags 0–7 and cumulative moving averages 01–07, adjusted for seasonality, day of week, holidays and meteorology, stratified by sex, age and season. Mean PM_{2.5} 32.0 µg m⁻³. PM_{2.5}, arsenic, cadmium, chromium, lead, manganese, nitrate and ammonium were positively associated; cadmium showed the largest association while nitrate and ammonium acted at delayed lags consistent with secondary inorganic aerosol. Preprint, not peer reviewed; no effect sizes or intervals in the abstract; constituents are highly collinear and single-pollutant time-series models will attribute shared variance arbitrarily among them. Fever-clinic attendance is also a behavioural endpoint sensitive to concurrent epidemic activity.

doi: 10.21203/rs.3.rs-10211219/v1

Reproductive & developmental

4 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

GBD 2023 India Stillbirth Collaborators 2026 Lancet Reg Health Southeast Asia

State-level stillbirth rates in India at ≥22 and ≥28 weeks, GBD 2023

First GBD estimation of stillbirth rate under the WHO ICD-11 ≥22-week definition alongside the conventional ≥28-week cutoff, disaggregated to 31 Indian geographic units. 122 data sources and 2,684 location-year observations were modelled by spatiotemporal Gaussian process regression on the stillbirth-to-neonatal-mortality ratio, anchored on GBD 2023 fertility and all-cause neonatal mortality, with 95% uncertainty intervals and a secondary analysis against SDG3 indicators. India carries the largest global stillbirth burden at both cutoffs, and the ≥28-week convention materially understates it. Air pollution enters only through GBD's risk-factor architecture, so no PM exposure–response is estimated here. Carried because state-level disaggregation is the resolution at which a PM–stillbirth analysis could be attempted, and because it defines the denominator such an analysis would use.

doi: 10.1016/j.lansea.2026.100819

Guo et al. 2026 Reprod Toxicol*Environmental pollutants and testicular toxicity: basic evidence and translational prospects*

Narrative review of testicular-level evidence for microplastics, bisphenols and air pollution, against a reported >50% decline in global sperm concentration over five decades. The PM-relevant claims: **spermatogenesis stage I, 70–90 days before semen collection, is nominated as the PM_{2.5} vulnerability window**, with metal constituents of traffic-derived PM_{2.5} as the primary toxicity driver. Separately, microplastics are reported in human testicular tissue with a gut-dysbiosis → immune activation → blood–testis barrier pathway, and BCAT1-mediated ferroptosis is offered as a bisphenol injury mechanism. Limitation: no systematic search or quality appraisal, so the window and constituent claims inherit whatever selection bias the cited primary literature carries. Read alongside Russo et al. — convergent on timing, but not independent evidence for it.

doi: 10.1016/j.reprotox.2026.109310

Li et al. 2026 Environ Pollut*Prenatal NO₂ and stillbirth in 3.98 million Iranian mother–baby pairs*

3,982,304 pairs, 2013–2018, pregnancy-average NO₂ from satellite-based gridded estimates averaged within 10 km of the delivery hospital, analysed under both the Iranian (≥22 weeks) and WHO (≥28 weeks) stillbirth definitions. Per 10 µg/m³: **OR 1.206 (1.192–1.220) and 1.106 (1.088–1.125)** respectively, essentially unchanged on adjustment for PM_{2.5} and ozone. The exposure–response is monotonic with *steeper slope at low concentrations*. A 10 µg/m³ target would avert 7,686 (7,242–8,127) stillbirths, an attributable fraction of 25.8%. Exposure assigned at the delivery hospital rather than residence is a substantial misclassification channel in a country with heavy referral flows, and the effect size is large enough to warrant scepticism. Co-pollutant, not PM.

doi: 10.1016/j.envpol.2026.128908

Russo et al. 2026 Paediatr Perinat Epidemiol*Critical windows of air-pollution exposure during spermatogenesis and IUI outcomes*

Secondary analysis of the Folic Acid and Zinc Supplementation Trial: 592 male partners in the Salt Lake City area contributing **1,223 intrauterine insemination cycles** over nine months. Residential exposure came from Community Multiscale Air Quality models, averaged over the 74-day spermatogenesis window before insemination and separately over mitosis, meiosis I–II, spermiogenesis and spermiation, with generalised linear mixed models adjusted for age, season, income and co-pollutants. Per 5 ppb across the full window, **O₃ RR 0.71 (95% CI 0.55–0.93) and NO₂ RR 0.73 (0.50–1.05)** for hCG+ pregnancy, concentrated in mitosis and meiosis; live-birth results were similar and no pattern emerged for pregnancy loss. **PM_{2.5} and SO₂ were less consistently associated and no numeric estimate was given.** Limitation: an infertility-treatment population is not the general population, residential modelling ignores occupational and commuting exposure, and co-pollutant models cannot cleanly separate a traffic mixture. The stage-resolved exposure definition is the methodological contribution worth borrowing.

doi: 10.1111/ppe.70185

Mechanistic toxicology**11 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

Asplund et al. 2026 Toxicol Sci*Concentrated ambient PM_{2.5} reversibly suppresses erythropoiesis and proresolving lipids*

Whole-body concentrated ambient particle exposure in mice with targeted splenic lipidomics. CAP suppressed erythropoietin, circulating reticulocytes and bone-marrow erythroid precursors, and reduced splenic iron and heme despite preserved splenic architecture — diminished erythrocyte turnover, not structural damage. Proresolving lipid mediators were broadly suppressed with **lipoxin A₄** most consistent, and its receptor ALX/FPR2 downregulated under prolonged exposure. **Return to filtered air normalised the lipid profile, LXA₄ and the erythroid parameters.** The recovery arm is what makes this the strongest design in the issue; the open question is whether the same reversibility holds after chronic rather than sub-chronic exposure.

doi: 10.1093/toxsci/kfag092

Deng et al. 2026 J Biochem Mol Toxicol*Baicalin rescue of PM_{2.5}-induced testicular injury via DKK1–Wnt/β-catenin*

Oral-nasal PM_{2.5} exposure at **750 µg m⁻³, 4 h/day for 45 days**, with baicalin gavage 1 h before exposure at 50 and 100 mg/kg. PM_{2.5} lowered testicular weight and organ coefficient, sperm count and serum testosterone and downregulated StAR, CYP11A1 and CYP17A1; baicalin restored these dose-dependently alongside DKK1–Wnt/β-catenin pathway markers. The exposure concentration sits one to two orders of magnitude above ambient even in heavily polluted settings, which is the standing weakness of this whole sub-literature — the pathway may be real, the dose is not.

doi: 10.1002/jbt.71057

Gao et al. 2026 Zhongguo Yi Xue Ke Xue Yuan Xue Bao*SQSTM1/p62-mediated mitophagy in PM_{2.5}-aggravated asthma, and dusquetide rescue*

House-dust-mite-sensitised C57BL/6 mice exposed to PM_{2.5}, assessed by whole-body plethysmography, HE/PAS histology, TOMM20–LC3B colocalisation and LC-MS/MS mitochondrial proteomics. PM_{2.5} raised airway hyperresponsiveness, inflammation and mucus and increased mitophagy signal; Mdivi-1 inhibition reduced inflammation; proteomics identified SQSTM1/p62 as the hub, and dusquetide suppressed both p62 and mitophagy while relieving hyperresponsiveness. Mdivi-1 is a poorly selective DRP1 inhibitor with known off-target effects on complex I, so the mitophagy-specificity of the rescue is not secured by that arm alone.

doi: 10.3881/j.issn.1000-503X.16948

Mishra et al. 2026 bioRxiv (preprint)*OMA1–DELE1 as a shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity*

Human PBMCs exposed to ultrafine PM, black carbon and polystyrene nanoplastics under matched conditions, profiled across NRF2, HIF-1α, PGC-1α, TFAM, OMA1, DELE1, mitochondrial ND1, complexes I–V, NF-κB, TNF-α and NLRP3, with PCA, hierarchical clustering and correlation networks. All three cause oxidative stress and mitochondrial dysfunction but with distinct kinetics: **UFPM gives an acute antioxidant/adaptive response, black carbon a chronic OMA1–DELE1 integrated stress response with inflammation, nanoplastics a low-grade persistent adaptation.** The comparative design under matched dosing is what makes this worth reading; it is a preprint, expression-level only, with no functional respirometry.

doi: 10.64898/2026.07.29.741662

Qi et al. 2026 J Hazard Mater*Urban PM_{2.5} worsens renal injury and fibrosis through a lung–kidney IL-6/ADCY3 axis*

Repeated intratracheal urban PM_{2.5} at an environmentally relevant dose combined with renal ischaemia–reperfusion injury in mice, plus tubular epithelial cell work. PM exposure worsened renal dysfunction, tubular injury, fibrosis and lipid-droplet accumulation; the proposed route is PM → systemic inflammation with alveolar macrophages as the likely IL-6 source → suppression of ADCY3–CREB–PGC1α → impaired fatty-acid β-oxidation, increased glycolysis, mitochondrial dysfunction and fibrosis. IL-6 blockade and ADCY3 overexpression each rescued, in vivo and in vitro. Limitation: the renal-IRI co-insult makes this a second-hit model, so the finding concerns PM aggravating existing injury rather than causing nephropathy; and intratracheal instillation again overstates deposited dose rate. The extra-pulmonary axis and the metabolic mechanism are both worth tracking.

doi: 10.1016/j.jhazmat.2026.143119

Sharma et al. 2026 J Toxicol Environ Health A*Dim light at night modulates the neuroimmune response to burn-pit emission inhalation*

32 male Swiss Webster mice housed under dark nights or dim light at night, exposed at week 4 to filtered air or burn-pit emissions (6.9 mg/m³ mass, 4.03×10⁶ particles/cc) for 4 h/day over 6 days. **No behavioural differences** in open field, Y-maze or elevated plus maze. Hippocampal *Il6*, *Tnfα*, *Il1β* and cortical *Il6* were decreased in LD-exposed mice, with reduced hippocampal *Bmal1*; lung and liver were unchanged. Small *n* per cell, a single sex, and a cytokine direction opposite to the usual inflammatory prior — reported as a circadian-interaction signal, not as an inflammation result.

doi: 10.1080/15287394.2026.2708884

Shi et al. 2026 J Appl Toxicol*PM_{2.5} aggravates airway inflammation and barrier disruption in COPD mice via RAGE-dependent PERK/eIF2α/ATF4/CHOP*

Cigarette-smoke-induced murine COPD model with subsequent PM_{2.5} exposure. PM worsened lung function, emphysema, inflammation, apoptosis and epithelial tight-junction damage, and activated the PERK/eIF2α/ATF4/CHOP endoplasmic-reticulum stress arm. The causal claim rests on two rescue arms: the RAGE inhibitor FPS-ZM1 and the ER-stress modulator 4-PBA each attenuated the phenotype, placing RAGE upstream of ER stress rather than parallel to it. Limitation: instilled PM at a dose the abstract does not state, a smoke-primed rather than naive background, and pharmacological inhibitors of imperfect selectivity — FPS-ZM1 in particular. The RAGE→ER-stress ordering is the novel part; the downstream inflammatory readouts are well-trodden.

doi: 10.1002/jat.70358

Silva et al. 2026 Environ Toxicol Pharmacol*PM_{2.5} organic extracts in 2D monolayers versus 3D spheroids of bronchial epithelium*

16HBE cells in paired 2D and 3D culture exposed to solvent-extractable organic fractions of well-characterised ambient PM_{2.5} from three sampling periods. **Concordance between formats is endpoint-dependent, not uniform:** WST-1 metabolic activity, ROS fluorescence and extracellular metabolite exchange respond more strongly in 2D, while pro-inflammatory cytokine secretion is more evident in spheroids. Cell death is necrotic rather than apoptotic in both. The practical implication for in vitro PM toxicology is that model choice silently selects which hazard you can detect — 2D favours direct cytotoxic and oxidative readouts, 3D favours responses that need spatial organisation and exposure gradients.

doi: 10.1016/j.etap.2026.105125

Wang et al. 2026 Ecotoxicol Environ Saf*Multigenerational reproductive toxicity of three vehicular PM types in C. elegans*

SRM diesel PM, WHTC diesel PM and WLTP gasoline PM, differing in morphology and elemental composition, all produced dose-dependent brood-size reduction, germ-cell apoptosis, lower sperm count and morphological abnormality in parents. Mechanistically: suppressed SPE-8/SPE-9/major sperm protein/vitellogenin, activated *ced-3/-4/-9*, disrupted *set-2*, *met-2*, *spr-5*. **Effects persist to F1–F2 and recover by F3–F4 — intergenerational carryover, not stable transgenerational inheritance**, which is the distinction most papers in this space elide. Nematode model; extrapolation to mammalian germline is speculative.

doi: 10.1016/j.ecoenv.2026.120609

Wu et al. 2026 (JHM) J Hazard Mater*TNF-α and EGF mediate endothelial activation downstream of PM_{2.5} alveolar injury*

3D alveolar–vascular barrier constructed from BEAS-2B epithelium, THP-1 macrophages and EA.hy926 endothelium, exposed to **100 μg/mL SRM2786 PM_{2.5} for 24 h**. PM_{2.5} entered epithelial cells by endocytosis, produced mitochondrial damage and apoptosis, and raised TNF-α, EGF, IL-8, TGF-α and IL-1β; extracellular TNF-α and EGF crossed the barrier and activated endothelial PI3K–Akt, with Nrf2 and CYP1A1 induction marking progressive oxidative stress. The model supplies a mechanistic route from local alveolar injury to systemic endothelial dysfunction. The single exposure concentration is far above any ambient equivalent and there is no dose–response, so the result establishes the pathway's existence, not its relevance at realistic exposures — the standard and unresolved weakness of this literature.

doi: 10.1016/j.jhazmat.2026.143204

Yibcharoenporn et al. 2026 ACS Omega*Metabolic profiling of PM-induced developmental toxicity in zebrafish embryos*

Zebrafish embryos exposed 4–50 hpf to fine and coarse PM at 500 and 1000 μg/mL, with untargeted LC-MS/MS metabolomics. The fine fraction alone produced a **22 h hatching delay** and disrupted pantothenate/CoA, retinol and riboflavin metabolism; riboflavin fell significantly at the higher dose. Glutathione responses diverged by size — reduced glutathione depleted under the fine fraction but *elevated* under the coarse — which is the most interesting result here because it argues against a single shared oxidative mechanism scaled by mass. The dose is the problem: 500–1000 μg/mL of suspended PM is three to four orders of magnitude above any inhalable ambient concentration, so this establishes a mechanism, not a risk. The authors' closing suggestion of vitamin supplementation is not supported by anything in the design.

doi: 10.1021/acsomega.6c00972

Occupational & indoor

9 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Cuerda Barcaiztegui et al. 2026 Atmos Pollut Res*Classroom ventilation management and user activities vs. indoor air quality, Madrid primary schools*

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue. A two-year classroom field study on ventilation management and occupant activity is directly on this watch's deployment axis and is the single highest-priority abstract re-check for the next run.

doi: 10.1016/j.apr.2026.103170

Kaur et al. 2026 NanoImpact*Workplace graphene: the material you make is more toxic than the air you breathe*

Pilot study pairing occupational exposure assessment in an operating graphene production facility with in vitro toxicology on the very material collected there. Stationary and personal samplers fed elemental carbon analysis, Raman microscopy-spectroscopy (which confirmed graphene in the collected PM), and viability testing in A549 and BEAS-2B cells. **Personal EC ranged 3–484 $\mu\text{g}/\text{m}^3$** , peaking during graphite powder handling. The dissociation is the finding: **workplace-collected PM was not cytotoxic at tested concentrations, while dispersions of as-produced liquid-phase-exfoliated graphene caused dose-dependent viability loss**. **Limitations: one facility, pilot scale, two cell lines, acute viability only**, and collected-sample doses may simply be below any threshold rather than genuinely benign. Nonetheless the design — sample the workplace, then test that sample rather than a pristine surrogate — is the coupling this field keeps asking for.

doi: 10.1016/j.impact.2026.100642

Makanjuola 2026 Research Square (preprint)*Personal PM exposure and spirometry among 563 Southwest Nigerian street sweepers*

Cross-sectional, non-smoking sweepers aged 18–60 from Ado-Ekiti, Ibadan and Ikeja; personal $\text{PM}_{2.5}$ and PM_{10} by handheld PMS5003 at breathing height; spirometry to ATS standard. Mean $\text{PM}_{2.5}$ **$103.55 \pm 72.03 \mu\text{g}/\text{m}^3$** and PM_{10} 105.09 ± 73.95 , roughly $20\times$ and $7\times$ the WHO 2021 annual guideline. $\text{PM}_{2.5}$ correlates inversely with FEV_1/FVC ($r = -0.14$, $p = 0.001$) but **the association is fully attenuated in the adjusted model** ($\beta = -0.0001$, $p = 0.286$). A bare PMS5003 without RH correction at these loadings is a real measurement-error problem the paper does not address, and $\text{PM}_{2.5} \approx \text{PM}_{10}$ is itself a red flag for the instrument. Reported for the exposure characterisation, not the health inference.

doi: 10.21203/rs.3.rs-9928829/v1

Nath et al. 2026 J Anaesthesiol Clin Pharmacol*Operating-room particulate: HEPA helps, cautery undoes it*

Portable monitoring of CO_2 , TVOC, formaldehyde, $\text{PM}_{2.5}$ and PM_{10} across operating rooms with and without HEPA filtration. HEPA rooms were significantly cleaner ($\text{PM}_{2.5}$ **19.78 ± 8.81 vs $29.83 \pm 4.56 \mu\text{g}/\text{m}^3$** , $p = 0.026$; PM_{10} 29.93 ± 12.96 vs 38.61 ± 6.89 , $p = 0.04$) while other pollutants were comparable. Electrocautery raised TVOC, $\text{PM}_{2.5}$ and PM_{10} ; ORs stayed well below the control area in winter ($\text{PM}_{2.5}$ 17.0 vs $36.66 \mu\text{g}/\text{m}^3$); average CO_2 was often high enough to matter for staff. **Limitations: small, single-centre, cross-sectional, with no personal sampling and no health endpoint**, and the HEPA and cautery comparisons appear to reuse overlapping room-groups, so the two effects are not cleanly separated. Included as a reminder that the highest-value indoor PM measurements are often minute-resolved and event-driven, which is where low-cost nodes are strongest.

doi: 10.4103/joacp.joacp_380_25

Niari et al. 2026 J Environ Health Sci Eng*Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid*

Thirty-six indoor air and 18 nasal lavage samples across salons and homes in Ardabil. Salon air carried **99.4 ± 4.1 versus $36.6 \pm 3.1 \text{MPs m}^{-3}$** at home, with nasal lavage $12.3\text{--}23.8$ versus $9.3\text{--}20.1 \text{MP mL}^{-1}$ and estimated daily exposure 572 ± 238 versus $212 \pm 109 \text{MPs/day}$. The nasal-lavage arm is the useful part — an internal-dose measurement rather than an inferred one — but the sample is small, no procedural blank is reported for a matrix acutely prone to airborne plastic contamination, and no polymer identification is described. Directionally credible, quantitatively soft.

doi: 10.1007/s40201-026-00999-9

Piedra et al. 2026 Front Public Health*Informal employment versus cantonal satellite pollution as respiratory-symptom predictors*

1,807 Ecuadorian workers across four sectors, ECRHS symptom instrument cross-referenced with Sentinel-5P retrievals; optimal spatial/temporal concentration percentile selected per outcome by AIC; Firth penalised logistic regression. **Informal employment dominates:** asthma aOR 12.56 ($5.09\text{--}32.87$), wheeze 6.78 ($1.83\text{--}27.89$), chronic cough 4.14 ($2.26\text{--}7.74$). After adjusting for cantonal altitude (itself protective, aOR 0.78 , $0.66\text{--}0.93$), NO_2 and SO_2 remain independent predictors of chronic phlegm and cough while ozone loses significance. Sentinel-5P carries no PM product, so the ambient arm is entirely gas-phase; cantonal assignment is coarse and the design is cross-sectional. The interesting move is using satellite retrievals as an occupational-surveillance covariate.

doi: 10.3389/fpubh.2026.1874197

Seyedrezaei et al. 2026 J Occup Environ Med*Ecological momentary assessment plus continuous IEQ sensing across home and office*

Nineteen office workers followed four months contributing **4,559 EMAs** of productivity, mood, stress and eustress–distress appraisal, against continuous CO_2 , TVOC, $\text{PM}_{2.5}$, temperature, humidity and noise, analysed with mixed-effects and multilevel mediation. Office work associated with higher productivity and greater eustress–distress coexistence, home with higher boredom and modestly lower stress; **home workspaces had poorer air quality**, and CO_2 and TVOC indirectly explained part of the mood and stress difference. $n = 19$ is very small for multilevel mediation, and the within-person design cannot separate the environment from everything else that differs between home and office — read the IEQ mediation as hypothesis-generating. The methodological interest is the pairing of dense subjective sampling with dense environmental sampling, which is the design template for sensor-linked wellbeing work.

doi: 10.1097/JOM.0000000000003833

Thies et al. 2026 Int J Popul Data Sci*The London Underground: very high $\text{PM}_{2.5}$, very little dose–response*

Two retrospective cohorts from the ONS Longitudinal Study of economically active adults living and working in London (1991, 2001, 2011 Censuses). Cohort 1 ($n = 7,343$) compared Underground commuters with above-ground rail users; Cohort 2 ($n = 4,094$) tested dose–response against externally linked cumulative LU $\text{PM}_{2.5}$ built from Transport for London data, $\text{PM}_{2.5}$ measurements and geospatial modelling. Outcomes were all-cause mortality (1991–2017) and cancer incidence (1991–2015), analysed with inverse probability weighting and the parametric g-formula. Most LU commuters were estimated to encounter hourly $\text{PM}_{2.5}$ above $100 \mu\text{g}/\text{m}^3$ ($8\text{--}10\times$ roadside), yet the survival deficit was **1.93% ($95\% \text{CI } 0.58\text{--}3.66$)**, cancer differences negligible, **no dose–response** emerged, and a hypothetical 50% $\text{PM}_{2.5}$ reduction barely moved survival. **Limitation: exposure is modelled, not measured on individuals**, and census-derived commute mode is a coarse, decadal proxy — non-differential misclassification of this size can flatten a real gradient. Either way, iron-rich mechanical-abrasion PM does not behave like ambient mass.

doi: 10.23889/ijpds.v11i5.3705

Tillery et al. 2026 Indoor Air*Indoor air nicotine, NO₂ and COVID-19 outcomes in a Great Plains tribal community*

COVID-19 Wayakta He? study: 290 multigenerational Cheyenne River Sioux Tribe households, each pair of adults at least 9 years apart in age, with one week of passive air-nicotine and NO₂ badges plus saliva serology. Geometric mean air nicotine 0.052 µg/m³ (detected in 40.1 % of homes); geometric mean NO₂ 10.5 ppb, range 0.65–138, consistent with propane cooking and heating. **OR 1.21 (1.01–1.51)** for prior infection with detectable air nicotine; **OR 3.69 (1.11–16.6)** for self-reported long COVID with NO₂ exposure. The second interval spans a factor of fifteen and rests on a self-reported outcome in a cross-sectional design with one week of exposure measurement standing in for a multi-year window — it is a direction, not a magnitude. The study's real contribution is that it measured the microenvironment at all: no ambient network in the Northern Plains resolves household combustion, and community-deployed passive samplers are the only instrument that reached it.

doi: 10.1155/ina/7749725

Burden, policy & mitigation**19 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (19 OF 19 RECORDS IN THIS SUBTOPIC)

Bu & Sun 2026 J Air Waste Manag Assoc*PM emissions from China's iron and steel industry: patterns, drivers, implications*

LMDI decomposition over 99 Chinese sites, separating stack from fugitive emissions into five process factors: steel output, energy intensity, waste-gas generation coefficient, stack concentration and fugitive emission factor. **National PM emissions fell 23.6% from 2011 to 2016.** Stack concentration was the dominant inhibitor (to –31,558 t in 2015–16); the fugitive factor flipped from +5,037 t (2010–11) to –23,320 t; the waste-gas coefficient pushed emissions up throughout (+19,516 t in 2015–16); energy intensity contributed under 25%. Regionally the picture inverts — South and Northwest China show annual increases, with Northwest fugitive emissions reaching 232% of the annual change. Limitation: the series ends in 2016, predating the ultra-low-emission retrofit programme entirely. The stack-versus-fugitive split is what matters for siting monitors near industry: the two have completely different spatial signatures.

doi: 10.1080/10962247.2026.2698603

Caikai et al. 2026 J Infect Dev Ctries*Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990–2021*

GBD 2021 across 204 countries, decomposing deaths, DALYs and age-standardised rates for respiratory infection and TB across nine risk factors by age, sex and SDI, with APC and AAPC trend estimation. Age-standardised mortality attributable to the traditional risks fell steadily — **air pollution –3.18% annually**, malnutrition –4.46%, alcohol –3.35% — while metabolic-risk-attributable deaths rose in absolute terms despite falling rates. Tobacco-attributable mortality was **3.6-fold higher in men**, and children under 5 remain the group most exposed to environmental risks. Air pollution is a GBD attributable fraction here, not a measured exposure–response, so this is a trend context record: the attributable burden is falling on rate, which the accompanying absolute counts do not support as a success narrative.

doi: 10.3855/jidc.21667

Galvin et al. 2026 J Air Waste Manag Assoc*Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers*

Emission factors developed under *representative cyclic operation* for three appliances — a residential retention-head liquid-fuel boiler, a residential modulating-condensing gas boiler and a two-stage commercial boiler — across B20, B100, renewable diesel, natural gas and a 20% hydrogen blend. PM measured by EPA Methods 5 and 202 to capture filterable and condensable fractions, supplemented with real-time instrumentation. Two results matter: **burner on–off cycling drives incomplete-combustion pollutants and is not represented in current inventory factors**, which derive from steady-state operation of larger plant; and **condensable PM can dominate total PM**, especially in advanced gas systems. An optical sensor sited near residential combustion is blind to the condensable fraction at the stack and sees it only once it forms — a real limitation for source attribution near buildings.

doi: 10.1080/10962247.2026.2657442

Gardner-Frolick et al. 2026 iScience*Double disproportionality: few emitters, and they sit next to marginalised people*

National analysis asking two questions jointly — which industrial subsectors have emissions concentrated in a few facilities, and whether those facilities are disproportionately sited near structurally marginalised groups. Emission disproportionality holds across pollutants and subsectors, and siting patterns diverge by settlement type: **oil and gas extraction disproportionately affects marginalised urban populations, mining and quarrying rural ones.** Urban Indigenous populations are strongly affected by both all-emitter and high-emitter siting, and low-income and Black populations are more likely to live near *clusters* of emitting facilities — a cumulative-hazard framing that single-facility analyses miss. **Limitations: proximity is not exposure** (no dispersion modelling, no monitoring), self-reported inventory emissions, and areal-unit demographics vulnerable to the modifiable areal unit problem. The cluster result is the one that should drive monitor siting.

doi: 10.1016/j.isci.2026.116827

Goyal et al. 2026 Nat Commun*Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's interstate trade network*

Integrated nutrient, water, greenhouse-gas, economic and interstate-trade accounting for Indian agriculture, comparing nitrogen-focused against water-focused crop-restructuring pathways. The nitrogen-focused pathway cuts **nitrogen surplus by 13.4%, water use by 18.6% and greenhouse gases by 8.7%** while holding baseline calorie production, and delivers about **4.6-fold larger reciprocal co-benefits** than the water-focused alternative, worth roughly **USD 1.19 bn** in avoided nitrogen-pollution socio-environmental cost. Limitation: PM is entirely implicit — reactive nitrogen is a dominant precursor of secondary inorganic aerosol over the Indo-Gangetic Plain, but no ambient PM_{2.5} response is modelled and no health endpoint is estimated, so the PM-relevant co-benefit is asserted through the nitrogen accounting rather than computed. Carried because agricultural NH₃ is the least-constrained term in South Asian PM source apportionment.

doi: 10.1038/s41467-026-75905-w

Gu et al. 2026 J Hazard Mater

Electro-ultrasonic coupled agglomeration for removal of fine particles from biomass combustion

Segmented continuous-flow bench platform combining bipolar charging (± 16 kV), a 300 V/400 Hz AC field and 16 kHz/142 dB ultrasound, with particle-trajectory simulation and SEM. The coupled field reached **86.4 % light transmittance**, against single-field results 33.1 and 67.2 percentage points lower, with the extinction coefficient down to 0.9 m^{-1} ; simulated cumulative particle displacement rose $>445\%$ as linear trajectories became three-dimensional helical paths, and SEM confirmed 15–20 μm agglomerates. The performance metric is the weakness: light transmittance and extinction coefficient are *optical* proxies, not gravimetric or number-resolved removal efficiency, so the headline “removal efficiency” is not directly comparable with any filtration literature that reports mass penetration. The submicron fraction claimed as the target is precisely the fraction an extinction measurement is least sensitive to.

doi: 10.1016/j.jhazmat.2026.143130

Han et al. 2026 AJPM Focus

Air quality index product use versus clinician discussion among US children with asthma

Questionnaire administered inside the Transcript/Omics for Precision Medicine in Asthma study, youth aged 6–21 recruited 2023–2025, with bivariate testing. **75% of caregivers were aware of an AQI or alert, 64% checked it before going outdoors and 73% reported a behaviour change** — but the delivery channel is consumer, not clinical or public: a phone weather app in 75% of cases, broadcast or social alerts 44%, websites 24%, county services only 14%. Only 41% had ever discussed air pollution with a clinician and 35% received poor-air-quality recommendations. Self-report with no objective behaviour measure and a single research cohort limit generalisability. The finding that matters for network design: **the operative last mile is a commercial weather app, and public monitoring agencies reach one family in seven.**

doi: 10.1016/j.focus.2026.100504

Hou et al. 2026 J Hazard Mater

Bimodal polysulfonamide fibrous aerogel for high-temperature $\text{PM}_{0.3}$ filtration

Co-electrospun aerogel combining thermally stable PSA/PAN/ SiO_2 coarse fibres with pure PSA fine fibres in a 3D curly network at **$>98\%$ porosity**, formed via moisture-induced phase separation and jet whipping from the mechanical mismatch between components. **$\text{PM}_{0.3}$ filtration efficiency 98.5% at ~ 84 Pa pressure drop, retained at 92.31% and 73 Pa after 12 h at 250°C** , with acid and alkali resistance and useful thermal insulation. The gap in the report is loading behaviour: no long-duration dust-holding capacity or efficiency-versus-loading curve is given, and that is what determines service life in a real industrial stack. Relevant to source-side mitigation rather than monitoring, and to the ultrafine fraction that mass-based networks under-weight.

doi: 10.1016/j.jhazmat.2026.143218

Kim et al. 2026 J Environ Manage

Full diesel-to-BEV replacement: where the benefit lands is not where the people are

CMAQ counterfactual in which all diesel vehicles are replaced by battery electric vehicles, evaluated with population-weighted concentration across urban core, urban periphery and rural strata — an explicit attempt to move policy appraisal off regional-average concentration change. Despite large on-road NO_x declines in the urban core and periphery, **annual daily-maximum 1-h ozone rose there through weakened NO titration** while falling in rural areas, so ozone PWC increases were concentrated where people live. **$\text{PM}_{2.5}$ moved the opposite way**: the largest concentration reductions occurred rurally, where low population density limited the exposure benefit. The result is a genuine spatial-equity tension inside a policy usually scored as unambiguously good. **Limitations: an idealised full-replacement scenario, one domain, one meteorological year**, no fleet-turnover or grid-emissions accounting, and no health-endpoint monetisation.

doi: 10.1016/j.jenvman.2026.130597

Kokol et al. 2026 Res Sq (preprint)

Artificial intelligence in the management of climate-change-related health issues in Africa

Scoping review over Scopus to October 2025, VOSviewer clustering a global against an African literature corpus. The divergence is the finding: global work clusters on environmental modelling, air-pollution mortality tracking and exposome integration, African work on supervised classification for infectious-disease triage and food security, with a “compute deficit”, data silos and power instability named as barriers. This is bibliometrics, not evidence — a VOSviewer map measures what Scopus indexed, and Scopus under-indexes African journals, so the gap is partly an artefact of the instrument. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-9910414/v1

Shin et al. 2026 J Occup Environ Med

Occupational disparities in $\text{PM}_{2.5}$ -related mortality under distributed lag models

Five years of daily all-cause mortality and $\text{PM}_{2.5}$ for Seoul, analysed with generalized additive plus distributed lag models and stratified by age, sex and occupation. A significant positive $\text{PM}_{2.5}$ –mortality association overall, with **blue-collar and elementary workers carrying substantially higher relative risks**. The policy argument — that concentration targets alone are insufficient and high-vulnerability strata need tailored management — is well posed, but the indexed abstract reports *no point estimates or intervals*, so the magnitude cannot be assessed and the record contributes no effect estimate. Occupation comes from death certificates, which misclassifies the late-career and retired.

doi: 10.1097/JOM.0000000000003831

Sidharaddi et al. 2026 Research Square (preprint)

Electrodeposited copper nodule films as electrostatic particulate collectors

Copper nodules electrodeposited onto steel; XRD confirms polycrystalline FCC, SEM a nodular microstructure, AFM a non-monotonic dependence of grain size ($86.5 \rightarrow 224.6 \text{ nm}$) and roughness ($23.4 \rightarrow 76.2 \text{ nm}$) on bath concentration before both fall again. The 0.075 M film (721 nm thick) gave **88.78% removal of dhoop smoke and an air-purifying factor of 54.12 over 405 s**, exceeding a stainless-steel discharge rod. Chamber test with a single combustion source, no reference-grade particle instrument, no size resolution and no ozone measurement — which for a corona-type collector is the omission that matters. Included as an engineering-control data point, tier C.

doi: 10.21203/rs.3.rs-10126181/v1

Slavik et al. 2026 Environ Challenges

Objective versus subjective wildfire-smoke exposure and parental protective behaviour

2,086 parents in wildfire-prone western US and Canada, surveyed and linked to 3.5 years of smoke-monitoring data (2020–2023). Perceived exposure (sight or smell) tracks measured exposure but predicts behaviour far better: $b = 1.11$ (**0.92–1.30**) **additional protective actions for perceived exposure versus $b = 0.23$ (0.06–0.40) per $10 \mu\text{g}/\text{m}^3$ of measured $\text{PM}_{2.5}$** . The measured-exposure relationship is non-monotonic — rise, plateau, rise. Self-reported behaviour is the obvious weakness and recall is correlated with perception by construction, which inflates the contrast. Even discounted, the direction is a design constraint on any public-facing sensor network.

doi: 10.1016/j.envc.2025.101362

Wan 2026 Econ Hum Biol*Place-based ecological restoration and self-rated health: quasi-experimental evidence*

Six waves of the China Family Panel Studies against the staggered prefecture-level rollout of China's national ecological restoration pilot, two-way fixed-effects DiD. **+0.1697 points of self-rated health on a five-point scale**, no differential pre-trends in event study, robust to placebo tests, Goodman–Bacon decomposition, propensity-score matching, double machine learning, IV, and adjustment for overlapping environmental policies. Mediation runs through lower PM_{2.5}, greater greenness and better visibility alongside economic and psychosocial channels. The unusually thorough robustness battery is the reason to take it seriously; the mediation analysis is correlational and the PM_{2.5} share is not separately identified.

doi: 10.1016/j.ehb.2026.101636

Wojcik et al. 2026 ATS Sch*An air pollution curriculum for pulmonary and critical care fellows*

Kern six-step curriculum development across five domains — composition, health effects, *monitoring*, risk mitigation and environmental justice — with a 20-item knowledge scale built by modified Delphi and delivered as pre/post assessment over one academic year with spaced repetition. Knowledge rose in all five domains ($p < 0.001$). Single programme, no control arm, self-selected respondents and a knowledge rather than behaviour endpoint, so the effect size is uninterpretable; it is included because the explicit inclusion of *monitoring* as a clinical competency domain is a small but real indicator of where sensor-derived exposure data is expected to land clinically.

doi: 10.1093/atsscholar/aapag039

Yoon et al. 2026 Med J Aust*Heat versus bushfire-smoke PM_{2.5} across four arms of the Australian health system*

National ED presentations, inpatient admissions, MBS primary-care claims and PBS prescriptions, 2014–2020, for cardiovascular, respiratory and mental-health conditions, supporting Australia's first National Climate Risk Assessment. **Heat loads the front door** (ED RR 1.033 per °C for respiratory, 1.023 for mental health) while **PM_{2.5} loads admissions and prescribing** (RR 1.014 per 1 µg/m³ at 1-day lag for both cardiovascular and respiratory admissions; primary-care mental-health visits RR 1.049, 1.040–1.058). Mental-health services are sensitive to both exposures across every domain — the least-anticipated result and the one with the weakest mechanistic account. Ecological in exposure assignment.

doi: 10.5694/mja2.70261

Zaid et al. 2026 J Mater Sci*Fibrous aerosol filtration: theory, CFD assumptions, metrology and fabrication*

Review integrating single-fibre capture mechanisms and most-penetrating-particle-size theory with electrostatic charge generation, transport and stability; 3D reconstruction of filter geometry; CFD, particle tracking, deposition and clogging simulation; and electrospinning, meltblowing, bubble electrospinning and supersonic solution blowing. The section worth reading is the critique of the homogeneity and steady-state assumptions that most published simulations inherit, alongside an explicit treatment of instrumentation limits and measurement uncertainty in filtration testing. Review, no new data; the material–process–performance framing is the organising contribution.

doi: 10.1007/s10853-026-13269-8

Zhang et al. 2026 J Hazard Mater*Particle-number emissions from a hybrid electric vehicle across cold and warm starts*

A China-6 compliant HEV tested over real-driving-emissions and RTS95 cycles with fresh and bench-aged aftertreatment. The particle-number size distribution was trimodal and dominated by an accumulation mode peaking near **90 nm**, with most emission from urban and highway segments and **engine-start events, cold and warm alike, as the dominant source**. Counter-intuitively, catalyst ageing *improved* overall PN removal: oxidation efficiency fell but gasoline particulate filter filtration rose as an ash layer established. Optimising warm-start fuel enrichment cut PN further on the aged system. Limitation: one vehicle, so this characterises a configuration, not a fleet. Directly relevant to near-road sensing — a 90 nm mode driven by discrete start events is largely invisible to mass-based instruments.

doi: 10.1016/j.jhazmat.2026.143156

Zhang et al. 2026 (preprint) medRxiv (preprint, not peer reviewed)*Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities*

Counterfactual policy evaluation for New York City, Los Angeles and Atlanta, 2005–2019. Daily PM_{2.5}, criteria gases, 1-in-3-day PM_{2.5} components and PMF-resolved PM sources at six monitoring sites were used to construct the concentrations that would have occurred without the selected EGU, mobile-source and port policies; city-site-specific multi-pollutant Poisson models then linked observed concentrations to respiratory emergency-department visits and hospitalised cardiovascular ED visits, and the fitted models were applied to both concentration series. **Median PM_{2.5} reductions of 27–62% across cities, and 7.1% (95% UI 5.4–8.8%) of cardiorespiratory visits averted**. Limitation: not peer reviewed; the counterfactual is a modelled construct whose uncertainty is not propagated into the health interval, and multi-pollutant Poisson models over highly correlated criteria pollutants cannot cleanly assign the benefit to PM. The source-resolved PMF layer is the part worth following.

doi: 10.64898/2026.08.04.26359640

Other clinical endpoints**6 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (6 OF 6 RECORDS IN THIS SUBTOPIC)

DeAngelis et al. 2026 Soc Sci Med*Residential context and epigenetic ageing: a null for air pollution*

Health and Retirement Study participants providing venous blood in 2016 ($n = 3,187$, mean age 69), with multigroup structural equation models decomposing paths between a latent epigenetic-age-acceleration variable (GrimAge, PhenoAge, DunedinPoAm38) and five 2010 residential features: racial clustering, air pollution, greenspace, household income and perceived disorder. White respondents in areas with high White clustering showed lower EAA, about 25% of it accounted for by higher area household income; **among Black participants essentially no residential path reached significance, and air pollution was not an independent predictor in either group**. Limitations: **single-timepoint methylation**, exposure indexed to areal units six years before assay, and a null that is as consistent with measurement error in the area-level pollution term as with absence of effect. Carried as a negative result: coarse area-level pollution proxies are not resolving what individual-level exposure assessment might.

doi: 10.1016/j.socscimed.2026.119635

Jung et al. 2026 Ann Rheum Dis*PM_{2.5} and rheumatoid arthritis disease activity and flare*

Prospective cohort at a South Korean tertiary centre, 2021–2024: 1,070 patients, 12,583 outpatient visits, monthly mean concentrations of six pollutants, generalised estimating equations adjusted for demographics, serology, medication, socioeconomic and meteorological covariates, plus a within-patient case-crossover sensitivity analysis on daily concentrations. **PM_{2.5} alone was associated with flare (adjusted OR 1.113, 95% CI 1.017–1.218)** and with higher DAS28-CRP, CDAI, and tender and swollen joint counts; the association was stronger in women and non-smokers, and the case-crossover arm reproduced it for cumulative exposure beyond two weeks. Single centre, monitor-assigned exposure; the two-design agreement is what makes it convincing.

doi: 10.1016/j.ard.2026.06.018

Lees 2026 Research Square (preprint)*PM_{2.5} component mixtures and inpatient hospitalisation in chronic kidney disease*

2,511 *All of Us* participants with ICD-coded CKD, 19,656 hospitalisations over 22,448 person-years (2010–2019); NASA SEDAC estimates of organic and elemental carbon, ammonium, nitrate and sulfate linked at 3-digit ZIP; weighted quantile sum regression inside a negative binomial model. Median PM_{2.5} was only 4.4 µg/m³. **IRR 1.31 (0.99–1.73, *p* = 0.06) per quartile of the WQS index — formally null, with 93% of the mixture weight on nitrate**, 4% each on ammonium and elemental carbon, and zero on organic carbon and sulfate. WQS weights are unstable at this sample size and 3-digit ZIP assignment is very coarse. The nitrate concentration of weight is the hypothesis worth carrying, not the point estimate.

doi: 10.21203/rs.3.rs-10438586/v1

Opperbeck et al. 2026 Environ Int*Exposome-wide search for epigenetic ageing — and how much is still missing*

FinnTwin12: 847 participants measured at ages 12, 14, 17 and 22, with 186 unique exposures spanning lifestyle, green environments, air pollutants and demographics, analysed by exposome-wide association plus knockoff boosted trees, sNPLS and Boruta to allow non-linearity and interaction. Epigenetic age at 22 (PCGrimAge, DunedinPACE) was **~28% explained** (*R*² 25.7% and 30.8%). Smoking, alcohol, youth unemployment and green-space size predicted accelerated ageing; tree cover, vegetation index, neighbourhood age structure and **aerial black carbon predicted decelerated ageing** — a sign reversal that the authors do not resolve and that almost certainly reflects urban socioeconomic confounding rather than a protective effect of BC. Twin modelling attributes the missing variance mainly to environmental factors unshared between co-twins. **Limitations: one cohort, modest *n* for 186 exposures**, and BC assigned at residence. Read for the missing-exposome decomposition, not the BC coefficient.

doi: 10.1016/j.envint.2026.110430

Sassano et al. 2026 Environ Health Perspect*Household solid-fuel use and kidney-disease mortality in the Golestan Cohort*

50,045 adults aged 40–75 enrolled 2004–2008 in northeastern Iran and followed to April 2023, with validated fuel-use questionnaires and Cox models; 262 kidney deaths over **724,064 person-years**. Per 10 additional years of household biomass use for cooking or heating, **HR 1.20 (95% CI 1.04–1.37)** for kidney-disease mortality; kerosene 1.09 (0.95–1.24) and gas 1.00 (0.86–1.16) were null. **The biomass effect splits on ventilation** — 1.19 (1.06–1.34) without a chimney against 1.06 (0.95–1.18) with one, *P*_{diff} = 0.025 — which is the strongest evidence in the paper that the exposure is the smoke rather than a correlate of poverty, though confounding by socioeconomic position is not fully excluded. **No PM measurement of any kind**: fuel type and duration are the entire exposure model, and the ventilation interaction is precisely the variable no ambient network can observe.

doi: 10.1021/EHP.6c00520

Yang et al. 2026 J Pain*Long-term exposure to multiple air pollutants and chronic pain in middle-aged and older Chinese adults*

China Health and Retirement Longitudinal Study, 2011–2020: **10,378 participants free of chronic pain at baseline, 8,047 incident cases over a mean 7.39 years** — an incidence approaching 78%, itself the study's most important number and a strong hint that the outcome definition is broad. Time-dependent Cox and accelerated failure time models were combined with weighted quantile sum regression over the five-pollutant mixture. PM_{2.5}, PM₁₀, SO₂ and CO were each associated with higher risk and earlier onset, and **SO₂ was the predominant mixture contributor**, not PM. Limitation: no interval estimates appear in the abstract, self-reported pain invites differential misclassification, and in a highly correlated mixture WQS weights identify the best-measured proxy as readily as the causal agent — read “SO₂ drives it” as a hypothesis.

doi: 10.1016/j.jpain.2026.106400

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Jahanshahi et al.	Long-term PM _{2.5} and NO ₂ and dementia onset in a low-pollution setting	<i>Int J Population Data Science</i>	Registry cohort + geospatial	PM _{2.5} , NO ₂ (5-yr moving average)	Northern Ireland
Lin et al.	Lung-brain axis and lung dysbiosis as a route from inhaled pollutants to epileptogenesis	<i>Genes & Diseases</i>	Critical review	Inhaled air pollutants (un-specified)	n/a
Cardiovascular & metabolic					
Leiser et al.	Ambient PM _{2.5} and preclinical re-modelling of cardiac structure and function	<i>JACC Adv</i>	Observational - cohort	PM _{2.5} and NO ₂ (spatio-temporal model, annual mean at residence)	North America
Li et al. (Environ Int)	PM _{2.5} constituents, hepatic fat and stiffness in youth, and effect modification by PNPLA3 genotype	<i>Environ Int</i>	Cross-sectional imaging	15 PM _{2.5} components, visit-year mean at residence	North America
Liu et al. (CCDC)	Long-term ozone and cardiovascular disease: warm-season vs annual-average exposure windows	<i>China CDC Wkly</i>	Review / synthesis	Ozone (co-pollutant; no PM estimate reported)	Global / multi-region
Sewor et al.	Null: personal HAP exposure and blood pressure in rural Rwandan children at 193 µg/m ³ median PM _{2.5}	<i>Environ Res Health</i>	Observational - cross-sectional	PM _{2.5} and black carbon (48 h personal, Ultrasonic Personal Aerosol Sampler)	Sub-Saharan Africa
Shi et al.	Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates	<i>Catheter Cardiovasc Interv</i>	Modelling / inventory	Unspecified PM	Multi-country
Zhao et al.	Ischaemic stroke burden and attributable risk factors in adults aged 20-54, 1990-2021	<i>Medicine (Baltimore)</i>	Burden estimation (GBD)	Air pollution as GBD risk factor (PAF)	Global / multi-region
Çulpan et al.	Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis	<i>Rev Environ Health</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ (long-term, per 10 µg/m ³)	Global
Reproductive & developmental					
GBD 2023 India Still-birth Collab.	State-level stillbirth rates in India at >=22 and >=28 weeks, GBD 2023	<i>Lancet Reg Health South-east Asia</i>	Burden estimation (GBD)	Risk-factor attribution incl. ambient and household PM	South Asia
Guo et al.	Environmental pollutants and testicular toxicity in male reproductive health: basic evidence and translational prospects	<i>Reprod Toxicol</i>	Review / synthesis	PM _{2.5} (traffic-derived metal constituents)	Global / multi-region
Li et al. (Environ Pollut)	Prenatal NO ₂ and stillbirth in 3.98 million Iranian mother-baby pairs	<i>Environ Pollut</i>	Observational - cohort	NO ₂ (satellite-based gridded, 10 km buffer); PM _{2.5} and O ₃ as co-pollutants	Middle East & N. Africa
Russo et al.	Critical windows of air pollution exposure during spermatogenesis and intrauterine insemination outcomes	<i>Paediatr Perinat Epidemiol</i>	Cohort (secondary trial analysis)	PM _{2.5} , NO ₂ , O ₃ , SO ₂ (CMAQ, 74-day spermatogenesis window)	USA
Respiratory & allergic					
Cantuaria et al.	Source-specific PM _{2.5} , NO ₂ , UFP and elemental carbon and incident COPD in two pooled Danish cohorts	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} , UFP, EC, NO ₂ (10-y time-weighted, address history)	Denmark
Haque et al.	Airborne PM and respiratory health in Bangladesh: narrative review	<i>Health Science Reports</i>	Narrative review	PM _{2.5} , PM ₁₀ , UFP, gaseous co-pollutants	Bangladesh
Hur et al.	Pollen modifies the association between suspended particulate matter and mortality across eight Kyushu cities	<i>Environ Pollut</i>	Time-series / case-crossover	SPM (daily, Japanese suspended PM)	Kyushu, Japan
Hussein et al.	PM ₁₀ and emergency visits for asthma exacerbation in Petaling Jaya, Malaysia	<i>J Glob Health</i>	Time-series / case-crossover	PM ₁₀ , daily mean, DLNM lag 0-7	Southeast Asia
Jiang et al.	Short-term PM _{2.5} , PM ₁₀ and SO ₂ and respiratory-infection hospitalisation burden in kidney transplant recipients	<i>J Cent South Univ Med Sci</i>	Time-series / case-crossover	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, O ₃ ; 7-day pre-symptom cumulative	China

First author	Focus	Journal	Design	Metric	Region
Orr et al. (EHP)	Wildfire-season PM _{2.5} and influenza / influenza-like-illness risk across six western US states	<i>Environ Health Perspect</i>	Distributed-lag time-series	Wildfire-specific PM _{2.5} (per 10 µg/m ³)	North America
Shukla et al.	Hourly ambient NO ₂ and emergency admissions for acute asthma, Birmingham and Solihull	<i>BMJ Open</i>	Distributed-lag time-series	NO ₂ (hourly, co-pollutant proxy)	Birmingham, UK
Solazzo et al.	Season-by-pollutant interaction on the nasal microbiota of healthy adults	<i>Hygiene & Environmental Health Advances</i>	Repeated-measures biomarker	Indoor TSP; outdoor PM, BC, benzene, CO	Italy
Wang et al.	PM _{2.5} constituents and daily fever-clinic visits: a time-series study in Jinshan, Shanghai	<i>Res Sq (preprint)</i>	Time-series	PM _{2.5} mass + As, Cd, Cr, Pb, Mn, nitrate, ammonium	Shanghai, China
Mechanistic toxicology					
Asplund et al.	Concentrated ambient PM _{2.5} suppresses erythropoiesis and splenic proresolving lipid mediators, reversibly	<i>Toxicol Sci</i>	Murine inhalation exposure	Concentrated ambient particles (whole-body)	United States (Louisville CAP facility)
Deng et al.	Baicalin rescue of PM _{2.5} -induced testicular injury via DKK1-Wnt/beta-catenin signalling	<i>J Biochem Mol Toxicol</i>	Murine inhalation exposure	PM _{2.5} , 750 µg/m ³ , 4 h/day, 45 days (oral-nasal)	China
Gao et al.	SQSTM1/p62-mediated mitophagy in PM _{2.5} -aggravated HDM asthma, and dusquetide rescue	<i>Zhongguo Yi Xue Ke Xue Yuan Xue Bao</i>	Murine inhalation exposure	PM _{2.5} (HDM-sensitised C57BL/6)	China
Mishra et al.	OMA1-DELE1 as the shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity	<i>bioRxiv (preprint)</i>	Experimental / toxicology	Ultrafine PM, black carbon, polystyrene nanoplastics (matched conditions)	South Asia
Qi et al.	Exposure to urban PM _{2.5} exacerbates renal injury and fibrosis via the lung-kidney axis through IL-6/ADCY3 signalling	<i>J Hazard Mater</i>	Animal + in vitro	Urban PM _{2.5} (repeated intra-tracheal)	China
Sharma et al.	Dim light at night reshapes the neuroimmune response to burn-pit emission inhalation	<i>J Toxicol Environ Health A</i>	Experimental / toxicology	Burn-pit emissions, 6.9 mg/m ³ mass and 4.03e6 particles/cc	North America
Shi et al.	PM _{2.5} exacerbates airway inflammation and epithelial barrier disruption in COPD mice through the RAGE-dependent PERK/eIF2α/ATF4/CHOP pathway	<i>J Appl Toxicol</i>	Animal experiment	PM _{2.5} (in-stilled)	China
Silva et al.	2D monolayers and 3D spheroids disagree by endpoint, not by magnitude, for PM _{2.5} organic extracts	<i>Environ Toxicol Pharmacol</i>	Experimental / toxicology	Solvent-extractable organic fraction of characterised ambient PM _{2.5}	Latin America
Wang et al.	Vehicular PM reproductive toxicity carries to F2 but not F3 in <i>C. elegans</i>	<i>Ecotoxicol Environ Saf</i>	Experimental / toxicology	SRM diesel PM, WHTC diesel PM, WLTP gasoline PM	China
Wu et al. (JHM)	TNF-α and EGF mediate endothelial activation downstream of PM _{2.5} alveolar injury in a 3D barrier model	<i>J Hazard Mater</i>	Toxicological (in vitro)	SRM2786 PM _{2.5} , 100 µg/mL, 24 h	Computational / laboratory
Yibcharoenporn et al.	Metabolic profiling of PM _{2.5} - and PM ₁₀ -induced developmental toxicity in zebrafish embryos	<i>ACS Omega</i>	Toxicological (in vitro)	PM _{2.5} + PM ₁₀ jointly (500, 1000 µg/mL suspensions)	Thailand (laboratory)
Occupational & indoor					
Cuerda Barcaiztegui et al.	Classroom ventilation management and user activities vs. indoor air quality in Madrid primary schools	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM (unspecified) + CO ₂ , indoor	Madrid, Spain
Kaur et al.	Workplace graphene aerosol: personal EC exposure paired with in vitro lung toxicity	<i>NanoImpact</i>	Exposure assessment + in vitro	Airborne graphene / elemental carbon	Graphene production facility (Europe)
Makanjuola	Street sweepers at 20x the WHO PM _{2.5} guideline, with spirometry that does not follow	<i>Research Square (preprint)</i>	Observational - cross-sectional	Personal PM _{2.5} and PM ₁₀ (PMS5003 handheld, breathing height)	Sub-Saharan Africa
Nath et al.	Operating-room PM _{2.5} /PM ₁₀ and the effect of HEPA filtration and cautery	<i>J Anaesthesiol Clin Pharmacol</i>	Environmental surveillance campaign	PM _{2.5} , PM ₁₀ , CO ₂ , TVOC, HCHO	India
Niari et al.	Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid	<i>J Environ Health Sci Eng</i>	Observational exposure study	Airborne microplastics (MPs/m ³); nasal lavage MPs/mL	Ardabil, Iran

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Piedra et al.	Informal employment dwarfs cantonal satellite pollution as a respiratory-symptom predictor	<i>Front Public Health</i>	Observational - cross-sectional	Sentinel-5P NO ₂ , SO ₂ , O ₃ (cantonal percentiles; no PM product)	Latin America
Seyedrezaei et al.	Ecological momentary assessment and continuous IEQ sensing across home and office workspaces	<i>J Occup Environ Med</i>	Repeated-measures biomarker	PM _{2.5} , CO ₂ , TVOC, T, RH, noise	United States
Thies et al.	London Underground commuting, cumulative PM _{2.5} and long-term mortality	<i>Int J Population Data Science</i>	Retrospective cohort	PM _{2.5} (metro, iron-rich; modelled cumulative)	London, UK
Tillery et al.	Indoor air nicotine, NO ₂ and COVID-19 outcomes in a Great Plains tribal community	<i>Indoor Air</i>	Cross-sectional survey	Indoor air nicotine + NO ₂ (1-week passive badges)	South Dakota, USA
Sensing, forecasting & instrumentation					
Adeniji et al.	Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria	<i>PLOS Glob Public Health</i>	Monitoring-record analysis	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ , hourly 2012-2023	Sub-Saharan Africa
Astaneh et al.	Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified), source-tagged tracers	USA
Barkjohn et al.	Air Sensor Network Analysis Tool: an R-Shiny application for sensor QA/QC and reference comparison	<i>Atmosphere</i>	Software / tool description	PM _{2.5} , O ₃ (low-cost sensor vs regulatory reference)	United States
Buettner et al.	PIA v3.0.0: automated quality assurance and analysis for PINE ice-nucleating-particle data	<i>Atmos Meas Tech</i>	Software / tool description	INP concentration; optical particle size discrimination	Germany (multi-site PINE)
Burke et al.	Polar-grid nonparametric trajectory analysis for near-source pollutant attribution	<i>Atmospheric Environment</i>	Methods / near-road field data	PM _{2.5} , BC, NO _x	St. Louis, MO, USA
Chung et al.	PurpleAir deployment trends, longevity and uncertainty across the United States, 2016-2025	<i>Sensors</i>	Network deployment + field evaluation	PM _{2.5} (PurpleAir PA-II)	United States (national)
Chyzykhov et al.	Particle-bound water and its bias on gravimetric PM mass	<i>J Hazardous Materials</i>	Laboratory metrology	PM constituents, artificial PM, ambient PM ₁ filters	Laboratory
Colarusso et al.	Mobile urban sensing: spatio-temporal observability analysis and optimization	<i>Internet of Things</i>	Method / algorithm	Reconstructed pollutant concentration maps	Italy
DeMarsh et al. (back-fill)	Community PurpleAir network versus regulatory monitors for PM _{2.5} characterisation in Fresno County	<i>Atmosphere</i>	Sensor co-location	PurpleAir PA-II PM _{2.5} vs regulatory FEM	North America
Ginsburg et al.	Transferability of US PurpleAir correction equations worldwide; regional recalibration	<i>Aerosol Science & Technology</i>	Multi-site co-location evaluation	PM _{2.5} (PurpleAir vs reference)	Global, 6 continents
Gomez et al.	Machine- and deep-learning calibration of low-cost PM ₁₀ /PM _{2.5} nodes against a T640X reference	<i>Sensors</i>	Calibration model comparison	PM ₁₀ , PM _{2.5} (LCS vs T640X)	Bogota, Colombia
Guo et al.	Mie-theory optical calibration plus aerosol liquid water separation for RH correction	<i>ACS ES&T Air</i>	Mechanistic correction framework	PM _{2.5} (nephelometer + OPC vs EPA reference)	United States (EPA CSN sites)
Gäbel & Hertig	Recalibration of low-cost O ₃ and PM _{2.5} sensors: linking practices to recent air sensor test protocols	<i>Atmos Meas Tech</i>	Field campaign / remote sensing	PM _{2.5}	Europe
Huang et al.	Metal ions and pH as controls on the optical properties of atmospheric brown carbon	<i>Atmos Pollut Res</i>	Chamber / laboratory	Brown carbon absorption spectra	Computational / laboratory
Jafari et al.	Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring	<i>IEEE Open J Intell Transp Syst</i>	Modelling / inventory	Unspecified PM	Europe
Jin et al.	Emissions, chemistry and health impact of aged wildfire smoke in Missoula, Montana	<i>Atmos Chem Phys</i>	Measurement campaign	Hourly PM _{2.5} , CO, NO _x , O ₃ and 75 speciated VOCs; GEOS-Chem	North America
Kumpika et al.	Joint humidity and sensor-aging correction for PMS7003 light-scattering sensors	<i>Environ Pollutants & Bioavailability</i>	Field co-location + chamber	PM _{2.5} (PMS7003 vs BAM)	Chiang Mai, Thailand
Kumpika et al.	Joint humidity and sensor-aging correction for low-cost light-scattering PM _{2.5} sensors	<i>Environ Pollut Bioavailab</i>	Mechanistic correction framework	PM _{2.5} (PMS7003 vs BAM reference)	Chiang Mai, Thailand
Lei et al.	Eight-year speciated PM ₁ record in Dublin: heating emissions vs trans-boundary transport	<i>Atmos Chem Phys</i>	Measurement campaign	PM ₁ (chemically speciated, ACSM + BC)	Europe

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Li et al.	Summertime ice-nucleating particle concentrations over the Eurasian-Arctic Seas	<i>Atmos Chem Phys</i>	Field campaign / shipborne	INP number concentration (immersion mode)	Arctic Ocean
Li et al. (Front Pediatr)	Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical + ambient exposure	<i>Front Pediatr</i>	Modelling / inventory	PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ (same-day and 0-7 d lag)	North America
Liu et al. (ACP)	From kappa to chi: PartMC evaluation of hygroscopicity-based mixing-state retrieval	<i>Atmos Chem Phys</i>	Modelling / inventory	HTDMA kappa distributions; mixing-state index chi (50-250 nm)	Global / multi-region
Magdaong et al.	Developing criteria and an algorithm for a low-cost IoT-based air quality sensor network for near-road air quality monitoring	<i>ISPRS Ann Photogramm Remote Sens Spat Inf Sci</i>	Method / algorithm	PM (low-cost sensor network siting)	Philippines
Masoom et al.	Long range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features of smoke particles	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	AOD; Angstrom exponent; single scattering albedo; volume size distribution	Multi-country
Nimmala et al.	Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling	<i>Environ Monit Assess</i>	Release-recapture field experiment	Bioaerosol (OFF-tagged pollen), high/low-volume samplers	Oliver Springs, TN, USA
Orr et al. (JMIR)	Feasibility and acceptability of a digital in-home air quality monitor in families of children with asthma	<i>JMIR Pediatr Parent</i>	Sensor deployment (pilot)	Indoor PM (consumer digital monitor)	North America
Oybek et al.	Chamber quantification of T/RH sensitivity across three low-cost sensor chemistries	<i>Am J Applied Science & Technology</i>	Controlled chamber factorial	PM _{2.5} (OPC), NO ₂ (MOx), CO ₂ (electro-chemical)	Laboratory
Qian et al.	AutoML multi-stage calibration for indoor low-cost PM _{2.5} sensors	<i>Atmospheric Measurement Techniques</i>	Chamber co-location, AutoML calibration	PM _{2.5} (indoor low-cost sensors)	Indoor chamber
Rehbein et al.	Source identification and triangulation from a low-cost sensor network in Ile-Ife	<i>Clean Air J</i>	Source apportionment	PM _{2.5} , PM ₁₀ , particle number concentration	Ile-Ife, Nigeria
Reis	Low-power sensor node with on-board quantised neural inference and adaptive LoRaWAN	<i>Sensors</i>	Simulation (system-level)	PM, gas, T/RH (embedded node)	Simulation
Rubio et al.	LoRa-connected micromobility pollution monitor for the Internet of Vehicles	<i>IEEE Consum Electron Mag</i>	Proof-of-concept deployment	PMx, CO ₂ , meteorology	Cartagena, Spain
Samburova et al.	Effects of western US wildfire smoke on urban air quality in northern Nevada	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} only (wildfire-influenced)	Nevada, USA
Sanders et al.	Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles	<i>Aerosol Sci Technol</i>	Chamber / laboratory	Hygroscopic growth factor, supermicron NaCl	Computational / laboratory
Silva et al.	Accuracy, applications and limitations of low-cost sensors for indoor air quality	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , CO ₂ (OPC and NDIR)	Global
Siu et al.	Disparities in fine particulate matter (PM _{2.5}) monitoring capacity in eastern Canada	<i>Environ Res Health</i>	Spatial analysis / GIS	PM _{2.5}	North America
Somnugpong	When seasonal refinement helps (and does not help) Stineman imputation of gappy air-quality series	<i>Environ Monit Assess</i>	Modelling / inventory	PM _{2.5} (UCI Air Quality; Beijing PM _{2.5})	China
Suryawanshi et al.	Spatiotemporal variation and hotspot identification of PM _{2.5} from low-cost sensors	<i>Int J Environ Res</i>	Metadata only (no abstract)	PM _{2.5} (low-cost sensors)	India (urban)
Tian et al.	When lidar-based CCN profiles fail: a heterogeneity index for vertically layered aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	CCN number concentration; lidar ratio and depolarisation ratio	North America
Tsai et al.	Vision-based vertical PM _{2.5} retrieval from a 5G-connected UAV with generative-AI interpretation	<i>IEEE Sens J</i>	Proof-of-concept deployment	PM _{2.5} (dark-channel-prior transmittance proxy)	Taiwan
Wang et al.	Traffic-induced environmental health assessment at intersections using roadside LiDAR	<i>Environ Monit Assess</i>	Modelling / inventory	Unspeciated PM	China
Yan	Review of machine-learning calibration for low-cost air quality sensor networks	<i>Sensor Review</i>	Narrative review	PM, O ₃ , NO ₂ (optical, electro-chemical, MOx, NDIR)	Global
Yang et al.	Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	PM _{2.5} ; PM ₁₀ ; total aerosol number	Europe

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Yaqoob et al.	Multi-city bi-pollutant ML calibration of PurpleAir and Alphasense OPC-N3	<i>IEEE Access</i>	Calibration model comparison	PM _{2.5} , PM ₁₀ (PurpleAir, OPC-N3 vs TEOM/FIDAS)	Antwerp, Oslo, Zagreb
Yee et al.	UCB-GLOBES: open-access GC/EI-MS spectral library for identified and unidentified organic aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	Semi-volatile organic aerosol (TMS derivatives, GC/EI-MS)	Global / multi-region
Yukhymchuk et al.	Low-cost particulate matter sensor performance under polluted conditions in Northeast China	<i>Aerosol Air Qual Res</i>	Sensor co-location	Low-cost PM sensor vs reference	China
Zhao et al.	WHA air-LSTM: composite morbidity-driven pollution index for next-day respiratory outpatient forecasting	<i>Biomed Environ Sci</i>	Modelling / inventory	Composite air pollution index (WHA air), city-specific ER curves	China
Zhivkov et al.	Distance-dependent calibration-transfer uncertainty in a hybrid reference/low-cost network	<i>Atmosphere</i>	Spatial cross-validation, operational network	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ (AirThings vs EEA reference)	Sofia, Bulgaria
Exposure assessment & modelling					
Baldo et al.	Chemical and mineralogical composition of Southern African mineral dust	<i>Atmos Chem Phys</i>	Laboratory metrology	Mineral dust (Si/Al, (Ca+Mg)/Al, K/Al, clay)	Southern Africa
Bhardwaj & Subramanian	Identifiability-aware source apportionment for sparse city-scale sensor networks	<i>arXiv (preprint)</i>	Source apportionment	PM (wind-conditioned lagged inverse problem)	Method (city-scale, synthetic + real)
Blanco-Villafuerte et al. (backfill)	Hybrid low-cost-sensor, satellite and CTM modelling of ambient PM _{2.5} in Peru	<i>Curr Environ Health Rep</i>	Machine-learning model	LCS + satellite AOD + CTM hybrid, 5 km daily	Latin America
Buhendwa et al.	Modeling the coupled PM _{2.5} -ozone system in China: chemical mechanisms, persistent model biases, and future perspectives	<i>Environ Pollut</i>	Review / synthesis	PM _{2.5} and O ₃ (chemical transport models)	China
Cai et al.	Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai	<i>Atmos Chem Phys</i>	Measurement campaign	Particulate organosulfates, FT-ICR MS	China
Ding et al.	Urban-rural balanced PM _{2.5} estimation corrects station-density bias	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} from AOD + reanalysis + ground	China, 2015-2021
Helbich et al.	Socioeconomic patterning of the urban external exposome depends on how you define it	<i>Environ Int</i>	Observational - cross-sectional	PM and NO ₂ within an 8-exposure external exposome	Europe
Hu et al.	Hourly 1-km seamless PM _{2.5} from ground, satellite AOD and WRF-Chem fusion	<i>Geo-spatial Information Science</i>	Machine-learning model	PM _{2.5} , AOD, WRF-Chem	China
Kim et al.	Wildfire emissions suppress downwind anthropogenic sulfate via OH scavenging	<i>Environmental Pollution</i>	CTM simulation + source attribution	PM _{2.5} sulfate	Northeast Asia (Siberian fire, Jul 2014)
Li et al.	Multi-media multi-pathway PAH exposure and carcinogenic risk under industrial, urban and rural scenarios	<i>Environ Monit Assess</i>	Physical exposure model	PAHs in air, water, soil, dust, food (BaP-eq)	China
Lu et al.	Moving-vehicle wakes and noise-barrier interaction in traffic emission dispersion	<i>Atmospheric Environment</i>	Metadata only (no abstract)	Traffic emissions / near-road dispersion	Not stated in deposit
Madalipay et al.	Establishing PM _{2.5} conversion factors for South Korea at provincial level	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} + PM ₁₀ jointly	South Korea
McCerery et al.	Arctic transport of North American and Canadian wildfire aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified) wildfire aerosol	Arctic / North America
Meena et al.	Transplanted lichen trace-metal accumulation and pigment response along a Jaipur urban gradient	<i>Environ Pollut</i>	Biomonitoring transplant	Zn, Pb, Cd, Cr, Cu, Ni by ICP-MS; chlorophyll / carotenoid	South Asia
Mushore et al.	Daily outdoor PM _{2.5} exposure and the road-edge effect across three sub-Saharan campaigns	<i>Atmos Environ X</i>	Measurement campaign	PM _{2.5} (personal, interpolated to surfaces)	Sub-Saharan Africa
Myat et al.	Ground, satellite and model assessment of PM _{2.5} dynamics and attributable mortality in Myanmar	<i>Environ Pollut</i>	Physical exposure model	PM _{2.5} speciated, TROPOMI HCHO/NO ₂ , GAM	Southeast Asia
Nguyen et al.	Systematic review of satellite-plus-ML PM _{2.5} prediction and explainable AI	<i>Engineering Research Express</i>	Systematic review (PRISMA)	AOD, TOA reflectance, Angstrom exponent -> PM _{2.5}	Global

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Parajuli et al.	Soil NO/HONO parameterisation choice and its effect on modelled O3 and PM	<i>Atmospheric Environment</i>	Model intercomparison	PM _{2.5} , O3, NO2 (CMAQ)	Continental United States
Pavlovic et al.	Aligning satellite Earth-observation capability, use case and capacity for air-quality health decisions	<i>J Air Waste Manag Assoc</i>	Mapping review	Satellite AOD-derived PM _{2.5} , trace gases	Global (GEO Health CoP)
Phalen et al.	Inhaled aerosol dosimetry update	<i>Inhal Toxicol</i>	Review / synthesis	Unspeciated PM; ultrafine	Multi-country
Pinakana et al.	Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools	<i>NPJ Clean Air</i>	Field campaign / remote sensing	PM _{2.5}	North America
Rai et al.	Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993-2023	<i>Environ Monit Assess</i>	Satellite remote sensing	MODIS AOD (2002-2023), Landsat LST	South Asia
Raptaki et al.	Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation	<i>Atmos Environ</i>	Metadata only (no abstract)	Secondary organic aerosol precursors (BTEX)	Heraklion, Greece
Roky et al.	Distance-resolved Pb dispersion and crop uptake from open battery-plate burning	<i>Environmental Pollution</i>	Dispersion model + soil sampling	PM _{2.5} , PM ₁₀ , Pb deposition	Kushtia, Bangladesh
Temu et al.	A six-domain framework for lichen biomonitoring including symbiont and microbiome response	<i>Environ Sci Pollut Res</i>	Literature review	Not applicable (framework)	Global / multi-region
Wandji Nyamsi et al.	Broadband AOD reconstructed from sunshine-duration records since 1900	<i>Atmos Chem Phys</i>	Physical exposure model	Broadband aerosol optical depth	Global (2,700 sites)
Wang et al.	Direct thermal vs emission-mediated pathways in heatwave-driven O3 and SOA increases	<i>Atmos Chem Phys</i>	Chemical transport model	SOA, O3 (BVOC and soil-NO pathways)	China
Xu et al.	Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities in winter	<i>Atmos Chem Phys</i>	Supersite observation	PM _{2.5} -bound nitrophenols and nitrocatechols	11 cities, China
Yadav et al.	Airborne endotoxin on PM ₁₀ in Agra: an 8-fold seasonal swing that PM mass does not track	<i>Research Square (preprint)</i>	Measurement campaign	PM ₁₀ mass; endotoxin (LAL); bacterial and fungal bioaerosol	South Asia
Yu et al.	Street-canyon increment to near-road NO2 quantified citywide with WRF-CMAQ-ADMS	<i>Environmental Pollution</i>	Chemical transport model	NO2 (traffic-related; canyon increment)	Hong Kong
Zhang et al.	H2O2-driven sulfate formation at air-water interfaces: stepwise mechanism and accelerated kinetics	<i>Environ Sci Technol</i>	Theory / simulation	Particulate sulfate formation rate	Computational
Zheng et al.	Spatially heterogeneous drivers of winter PM _{2.5} in China from synchronous observations and explainable machine learning	<i>Atmos Environ</i>	Machine-learning model	Winter PM _{2.5} , synchronous multi-site observation	China
Burden, policy & mitigation					
Bu & Sun	Particulate matter emissions from the iron and steel industry of China: patterns, drivers, and implications	<i>J Air Waste Manag Assoc</i>	Emission inventory / decomposition	PM (stack and fugitive)	China
Caikai et al.	Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990-2021	<i>J Infect Dev Ctries</i>	Burden estimation (GBD)	Air pollution as GBD risk factor (ASMR, ASDR)	Global / multi-region
Galvin et al.	Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers	<i>J Air Waste Manag Assoc</i>	Chamber emission characterisation	Filterable + condensable PM (EPA M5/M202) + real-time PM	United States
Gardner-Frolick et al.	Double disproportionality: concentrated PM _{2.5} emitters and marginalised siting in Canada	<i>iScience</i>	Spatial econometric model	PM _{2.5} and PM _{2.5} precursor emissions (NPRI)	Canada
Goyal et al.	Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's interstate trade network	<i>Nat Commun</i>	Modelling / inventory	Precursor emissions (N)	India
Gu et al.	Electro-ultrasonic coupled agglomeration for removal of toxic fine particles from biomass combustion	<i>J Hazard Mater</i>	Chamber / laboratory	PM (unspeciated) from biomass combustion	China
Han et al.	Use of air quality index products versus clinician discussion among US children with asthma	<i>AJPM Focus</i>	Cross-sectional survey	Self-reported AQI product use	North America
Hou et al.	Bimodal polysulfonamide fibrous aerogel for high-temperature PM0.3 filtration	<i>J Hazard Mater</i>	Materials / filtration	PM0.3 filtration efficiency, pressure drop	Computational / laboratory
Kim et al.	Spatial equity and cost-effectiveness of full diesel-to-BEV replacement	<i>J Environmental Management</i>	Chemical transport model	PM _{2.5} , O3 (population-weighted concentration)	Republic of Korea

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Kokol et al.	Artificial intelligence in the management of climate-change-related health issues in Africa: scoping review	<i>Res Sq (preprint)</i>	Scoping review	PM (unspecified), bibliometric corpus	Sub-Saharan Africa
Shin et al.	Occupational disparities in PM _{2.5} -related mortality under distributed lag models	<i>J Occup Environ Med</i>	Time-series	PM _{2.5} (daily, city-wide)	Seoul, South Korea
Sidharaddi et al.	Electrodeposited copper nodule films as electrostatic PM collectors	<i>Research Square (preprint)</i>	Experimental / toxicology	Dhoop-smoke particulate (chamber, unspecified)	South Asia
Slavik et al.	Perceived smoke beats measured smoke as a predictor of parental protective action	<i>Environ Chall</i>	Observational - cross-sectional	PM _{2.5} on smoke days (monitoring data) vs self-reported sight/smell	North America
Wan	Ecological restoration as a population health intervention, with PM _{2.5} as one mediator	<i>Econ Hum Biol</i>	Ecological	PM _{2.5} concentration (mediator alongside greenness and visibility)	China
Wojcik et al.	An air pollution curriculum for pulmonary and critical care fellows	<i>ATS Sch</i>	Educational intervention	PM (unspecified), curriculum content	USA
Yoon et al.	Heat vs bushfire-smoke PM _{2.5} across four arms of the Australian health system	<i>Med J Aust</i>	Ecological	PM _{2.5} (bushfire smoke) and temperature, short-term lags	Oceania
Zaid et al.	Fibrous aerosol filtration: where CFD assumptions and measurement uncertainty diverge from practice	<i>J Mater Sci</i>	Review / synthesis	Ultrafine particles; most penetrating particle size; pressure drop	Global / multi-region
Zhang et al.	Particulate emission risks of HEVs during cold-starts and warm-restarts: particulate number under fresh and aged aftertreatment systems	<i>J Hazard Mater</i>	Vehicle emission experiment	Particle number; particle number size distribution; sub-23 nm PN	China
Zhang et al. (preprint)	Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities	<i>medRxiv (preprint)</i>	Time-series / case-crossover	PM _{2.5} ; PM sources (PMF)	North America
Other clinical endpoints					
DeAngelis et al.	Residential context and epigenetic age acceleration: a null for air pollution	<i>Social Science & Medicine</i>	Cross-sectional (multistage)	Neighbourhood air pollution (1 of 5 contextual features)	United States
Jung et al.	PM _{2.5} and rheumatoid arthritis flare: 12,583 visits with a case-crossover sensitivity arm	<i>Ann Rheum Dis</i>	Observational - cohort	PM _{2.5} , PM ₁₀ and four gases (monthly means; daily for case-crossover)	East Asia (ex-China)
Lees	Null with a nitrate signal: PM _{2.5} component mixtures and hospitalisation in CKD	<i>Research Square (preprint)</i>	Observational - cohort	PM _{2.5} components OC, EC, NH ₄ , NO ₃ , SO ₄ (NASA SEDAC, 3-digit ZIP)	North America
Opperbeek et al.	Exposome-wide search for drivers of epigenetic ageing, and the size of the missing exposome	<i>Environment International</i>	Exposome-wide association	Aerial black carbon among 186 exposures	Finland (FinnTwin12)
Sassano et al.	Household solid-fuel use and kidney disease mortality in the Golestan Cohort	<i>Environ Health Perspect</i>	Prospective cohort	Household fuel type and duration (biomass / kerosene / gas), questionnaire	Middle East & N. Africa
Yang et al.	Associations of long-term exposure to multiple air pollutants with chronic pain among middle-aged and older adults in China	<i>J Pain</i>	Cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO (5-pollutant mixture, WQS)	China

Provenance and method

This rollout is an **aggregation only**. It pools `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over 2 August 2026–8 August 2026 via `PMRW_START/PMRW_END` and **re-queries no API**, so every record was screened and deduplicated in the daily issue that first carried it. Per-record provenance, source-query windows, exclusions and known metadata defects are documented in those daily issues and are not restated here. **Daily issues covered: 2, 3, 4, 5, 6, 7 and 8 August 2026 — seven of seven, no missed runs and no gap in the entry-date coverage. Known defects carried into this rollout, stated rather than hidden:** (i) one cross-issue duplicate, DOI 10.1080/26395940.2026.2616098, shipped on both 2 and 3 August, so 146 records span 145 unique DOIs; (ii) journal names are free text and three venues appear under two spellings each, which understates venue concentration in the register; (iii) 8 records carry no abstract from any checked source and are unassessed; (iv) several 2–7 August records were Consensus or Crossref backfills with creation dates outside the window and are excluded from trend claims but included in the pooled counts. Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption; **no meta-analytic summary is computed and none should be**. Trend and inflection statements are judgements over a seven-day window and are not statistical tests. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.